భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

Part 2.1: *p-n* Junction Diodes

EE2104: Semiconductor Device Fundamentals

EE, IIT Hyderabad

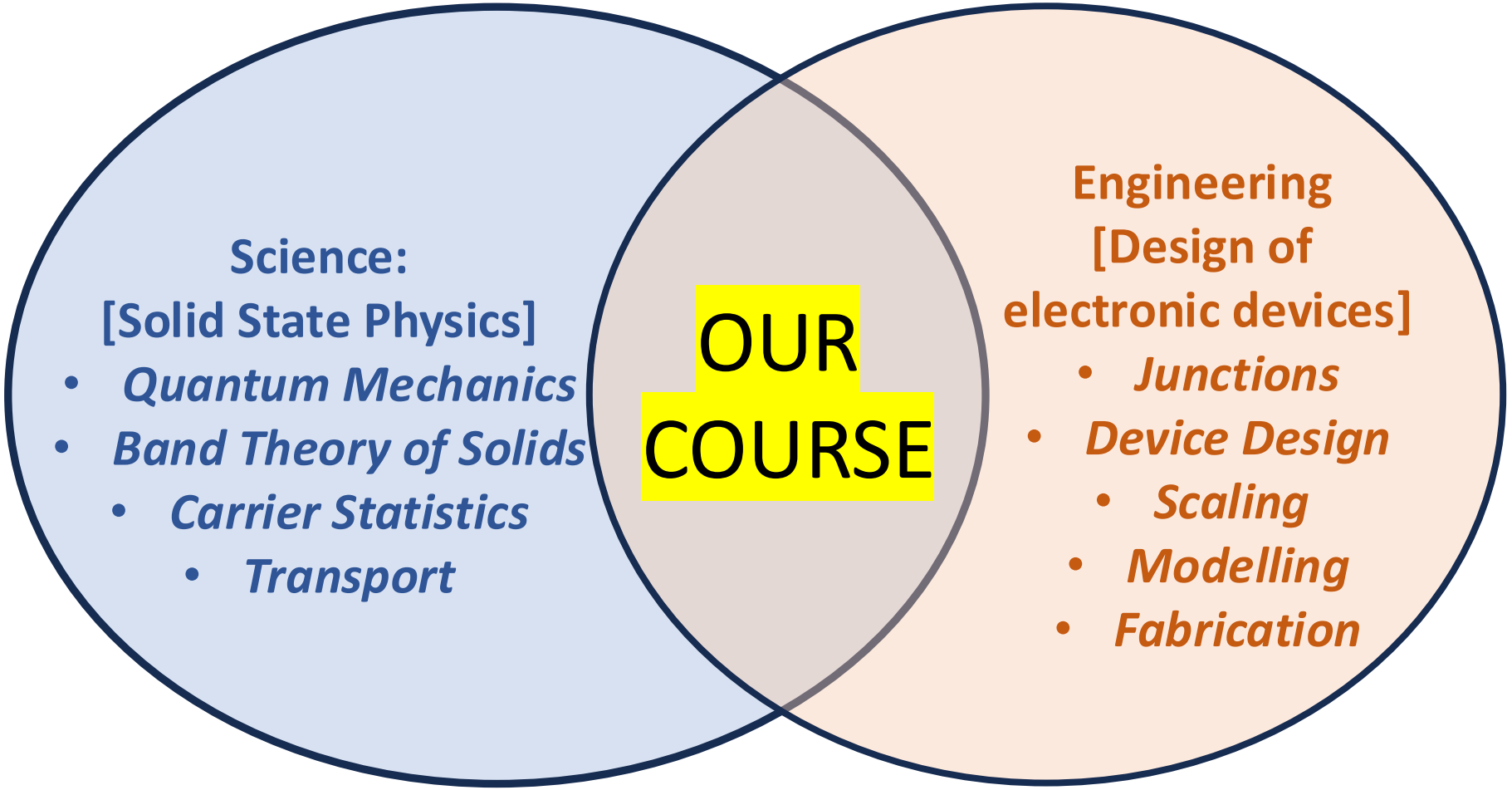
Shubhadeep B (EE507, shubhadeep@ee at iith)

Topics: *Recap, Diode Fabrication, Equilibrium Conditions, Non-equilibrium transport, Breakdown, Non-Idealities, CV Characteristics.*

AI Credits: Few schematics are made with Gemini ☺

Approach of our Course – And a reflection of this field

This area (Semiconductor devices) is very unique:
It combines pure science and extreme engineering:
That is what we **love** about this area!



This course requires experienced handholding, cramming from textbook at last moment WILL NOT work!

Appeal: Come to class & follow the material.

Promise: You will have a great time! ☺

Course Contents

Part 1: Semiconductor physics fundamentals (20-22 hrs)

- Intro/Motivation, Essentials of quantum mechanics (Wavefunction, Schrodinger's equation, particle in a potential well, tunneling) – Things you may know from +2/Modern Physics but with a new perspective.
- Band theory of solids (Qualitative formation of bands, Kronig Penney model, band gap, effective mass)
- Equilibrium carrier statistics (DOS, Fermi-Dirac Statistics)
- Non-equilibrium transport (Quasi-Fermi level, Transport: Drift/Diffusion, carrier continuity)

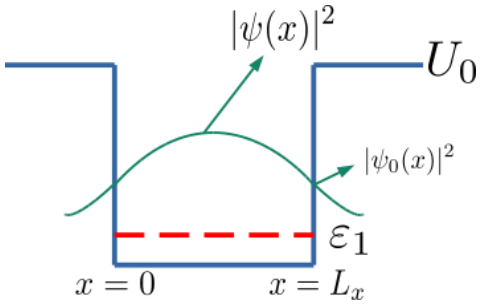
Part 2: Junctions and Devices (20-22 hrs)

- *Semiconductor-Semiconductor: pn* Junctions: band diagrams, ideal and non-ideal I-V characteristics, pn Junction C-V, *basics of heterojunctions and Schottky junctions* (if we have time).
- MOSCAPs (modes of operation, ideal and non-ideal CV characteristics)
- MOSFETs (long channel characteristics, short channel effects (SCE), device engineering to mitigate SCE)

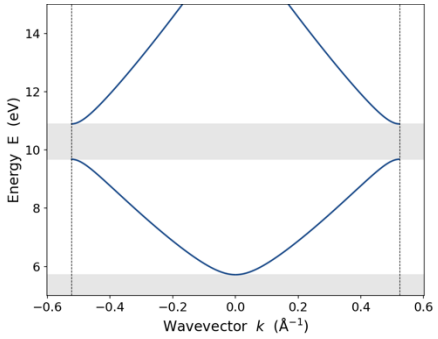
How do the semiconductor devices behave ? Semiconductor Devices → Electron Devices
Behavior → How many carriers are there ? How easily do they move around ?

[END OF PART -1 Credits: Prof. Oves Badami]

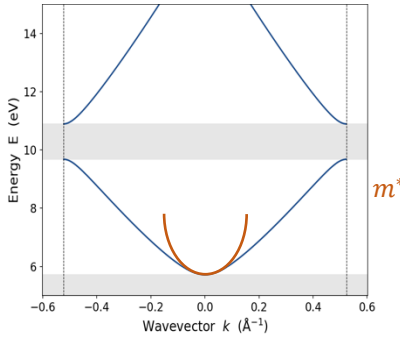
1.0 Quantum Mechanics
2.1 Infinite Potential Well
2.2 Finite Potential well



Chain of 1-D Finite Potential wells – Kronnig – Penney Problem

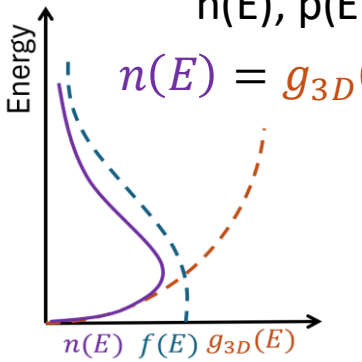


Approximate E-k → Parabolic E-k
Concept of Effective mass



$$m^* = \hbar^2 \left[\frac{d^2E}{dk^2}\right]^{-1}$$

Density of states + Fermi-Dirac statistics → n(E), p(E)

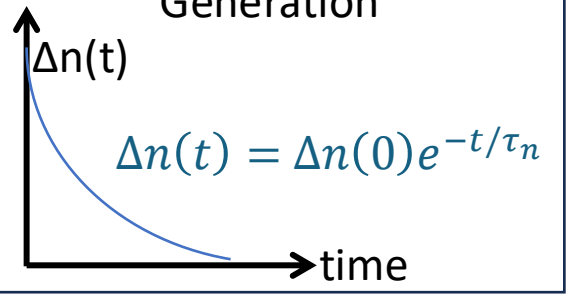


Non-Equilibrium case:
Applied Bias (Drift) &
Carrier injection (Diffusion)

$$J_n = qn\mu_n \boldsymbol{E} + qD_n \frac{dn}{dx}$$

$$J_n = \frac{\sigma}{q}\frac{dF_n}{dx}$$

How do we come back to equilibrium → Recombination and Generation

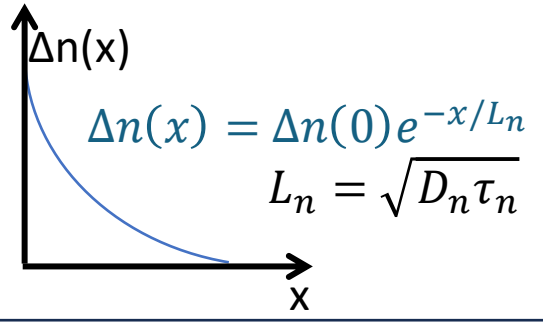


Putting everything together
Carrier continuity

$$\frac{\partial n}{\partial t} = \frac{1}{q}\frac{\partial J_n}{\partial x} + G - R$$

$$\frac{\partial p}{\partial t} = -\frac{1}{q}\frac{\partial J_p}{\partial x} + G - R$$

Application of Carrier continuity to a specific case → Diffusion Only



Application of these concepts to practical semiconductor devices → Diode, ~~BJT~~, MOS Capacitor, MOSFET

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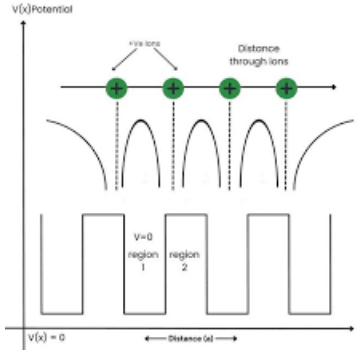
Approach of this course: ROADMAP

This Course: Semi-Classical Transport Modelling

Quantum mechanics

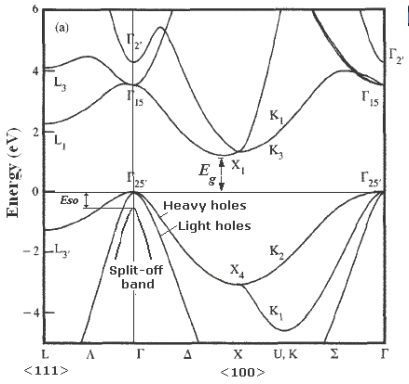
$$-\frac{\hbar^2}{2m}\nabla^2\psi + V(\boldsymbol{x})\psi = E\psi$$

Transport over large length scales >> λ_{dB} (De-Broglie Wavelength of e-)
e- can be approximated to a 'particle'



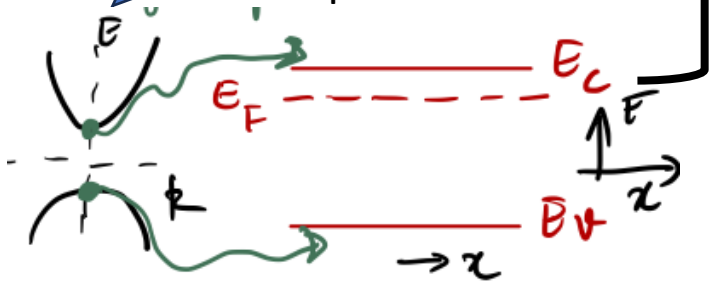
Approx 1

Particle Approx:
m* accelerated by 'E'



Approx 2

Near/Quasi Equilibrium:
Simple Band Picture

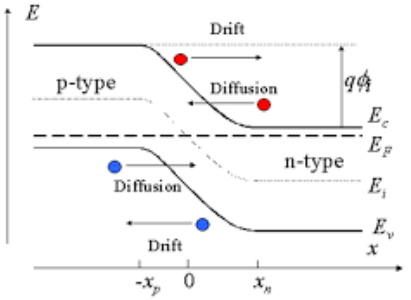


+ B.C.
V (end)

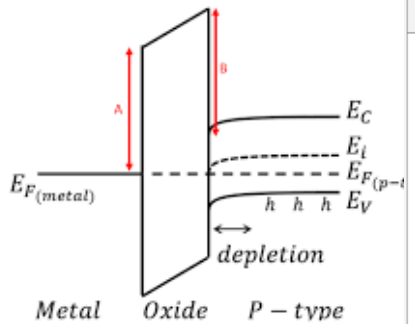
Semi-Classical
- Electrostatics: Poisson's Eqn
- Drift-Diffusion
- Carrier-Continuity

J_n, J_p

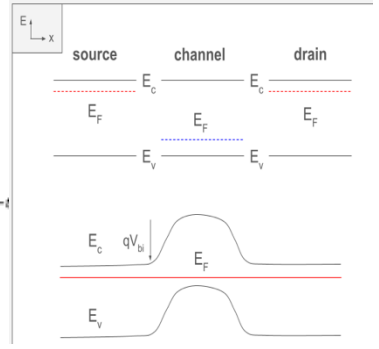
DIODE (p-n junction)



MOSCAP



MOSFET



Semiconductor Fundamentals (~20-22 hrs)

Junctions and Devices (20-22 hrs)

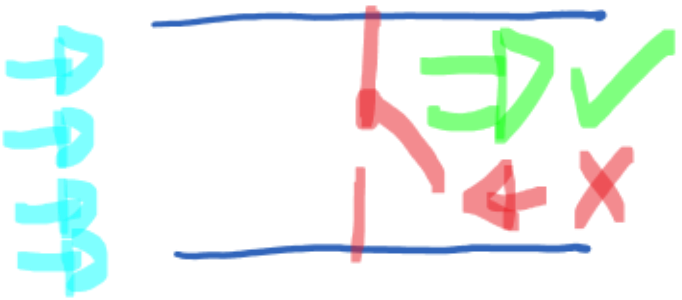
Essentials of Q Mech | Band Theory | Equilibrium carrier Stats | Non-Equ Transport

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Introduction to *pn* junction diode

What is a diode :- An "electronic valve"

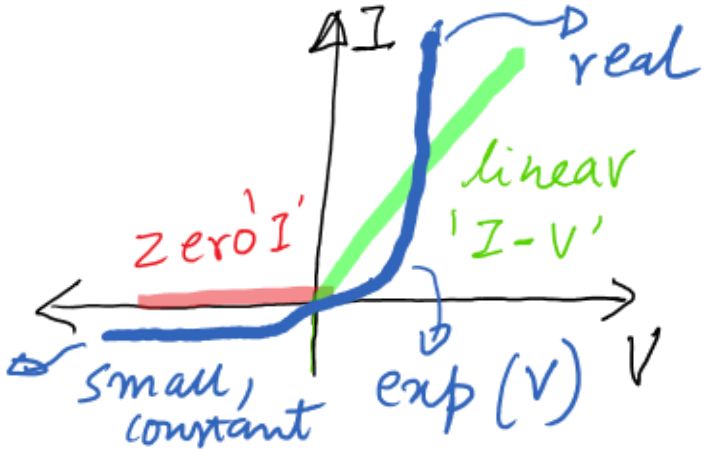
"WATER VALVE"



"ELECTRONIC VALVE"



expect vs. real I-V

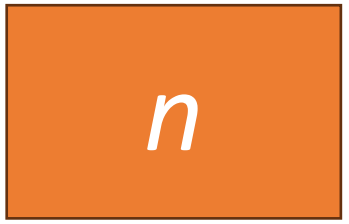


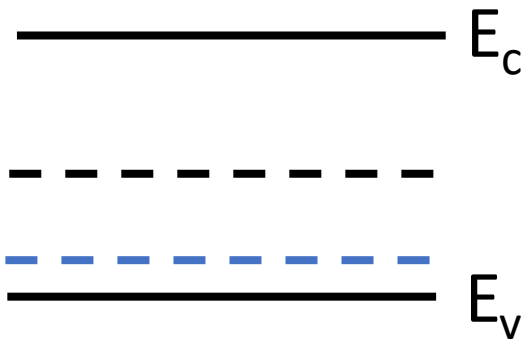
Questions to be answered in this module

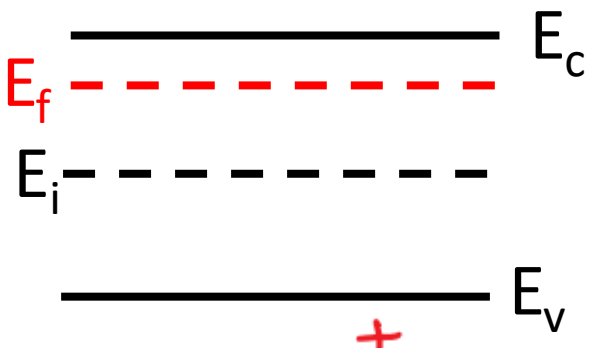
- What is structure & construction of a diode :- P-n Junc.
- Why / How rectifying behaviour? Equi vs. BIASING { non-equilibrium }
- Why "CHARGE/CURRENT INJECTOR" F.W bias $I \propto \exp(V)$ && REV BIAS $I = -I_0$ { rev sat } "Charge/Current REGULATOR" } ⇒ P-n Junc + MOS 2 pillars of Electronic Devices.
- Non-Idealities & "C-V char" ⇒ AC switching.

Useful Recap of Previous Concepts -1









carrier

$p_p \simeq N_a^- \{\text{acc. dopant}\}$

doping

$n_p = n_i^2 / p_p$

$p_p = N_v \exp\left[\frac{-(E_F - E_V)}{k_B T}\right]$

$\{\text{in } E_F\}$

$n_n = N_D^+$

$p_n = n_i^2 / N_D^+$

$n_n = N_c \exp\left[\frac{-(E_C - E_F)}{k_B T}\right]$

Important: p_p, n_n - MAJORITY carriers
: n_p, p_n - MINORITY carriers

* ASSUMPTIONS

↳ 100% Ionisation.

↳ $E_A - E_V$, $E_C - E_D$ } few $k_B T$

~ 26 meV @ R.T.

Remember n_n & p_p in terms of n_i & $E_F - E_i$

$n_n = n_i \exp\left[\frac{E_F - E_i}{k_B T}\right]$

$p_n = n_i \exp\left[\frac{E_i - E_F}{k_B T}\right]$

Useful Recap of Previous Concepts -1

1) Equilibrium conditions $\Rightarrow \boxed{J_{net} = 0}$ & $\frac{dE_F}{dx} = 0$ { no gradient in Fermi level }

2) Non-equilibrium $\Rightarrow \boxed{J_{net} \neq 0} \Rightarrow$ net flow of charge in one direction

Useful Recap of Previous Concepts -1

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causes "forces" of current flow:-

#1) Conc. gradient = DIFFUSSION

$\frac{dn}{dx}$ or $\frac{dP}{dx}$ { ∇n, ∇P

$J_{P,diff} = -q D_P \frac{dP}{dx}$; $J_{n,diff} = +q D_n \frac{dn}{dx}$

{ hole (electron) diff current opp. (same) as ∇.)

@ Equilibrium $J_{drift} = J_{diff} \Rightarrow \boxed{\frac{\mu_n}{D_n} = \frac{\mu_P}{D_P} = \frac{k_B T}{q}}$

Useful Recap of Previous Concepts -1

1) Equilibrium conditions $\Rightarrow \boxed{J_{net} = 0}$ & $\frac{dE_F}{dx} = 0$ { no gradient in Fermi level }

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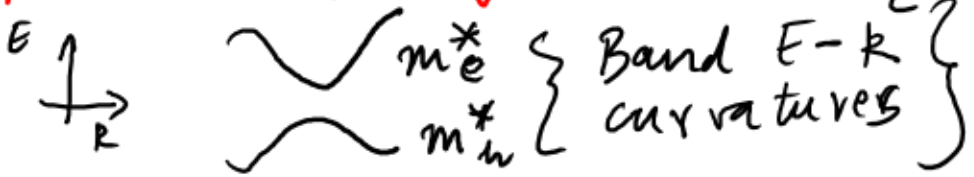
@ Equilibrium $J_{drift} = J_{diff} \Rightarrow \boxed{\frac{\mu_n}{D_n} = \frac{\mu_P}{D_P} = \frac{k_B T}{q}}$

#2) Electric Field ($\vec{\mathcal{E}}_{\phi}$) [DRIFT]

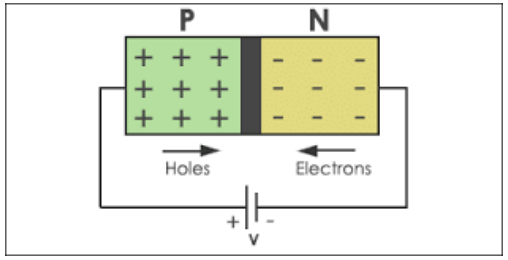
$J_{drift} = \sigma \vec{\mathcal{E}}_{\phi}$

$= [n q \mu_n + p q \mu_p] \vec{\mathcal{E}}_{\phi}$

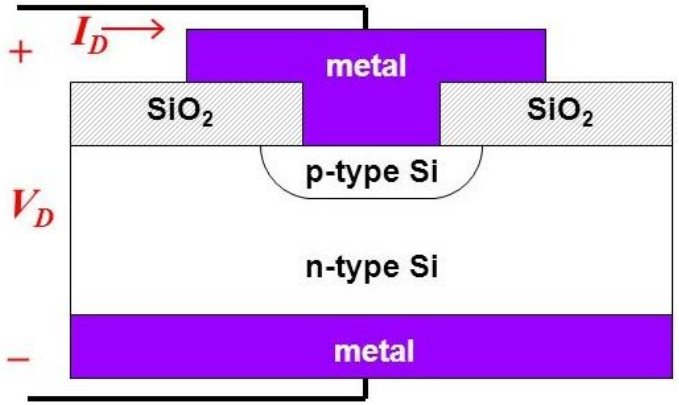
Recall origin of $\mu_n, \mu_P \simeq \frac{q \tau^*}{m^*}$



Small detour: pn - junction diode fabrication



p-n junction in our course



Physical structure (Cross-sectional view)

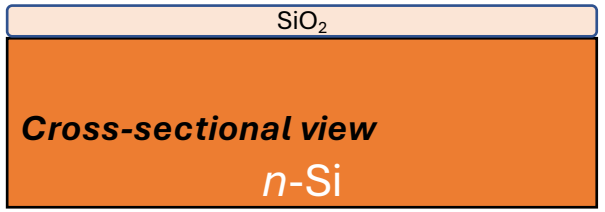
How do we Fabricate such a structure?

Step 1: Wafer Selection and Cleaning (RCA+bHF)



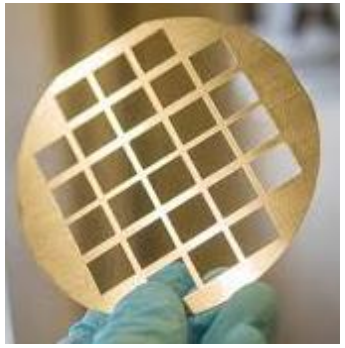


Step 2(Add): Wet/Dry Oxidation of *n*-Si

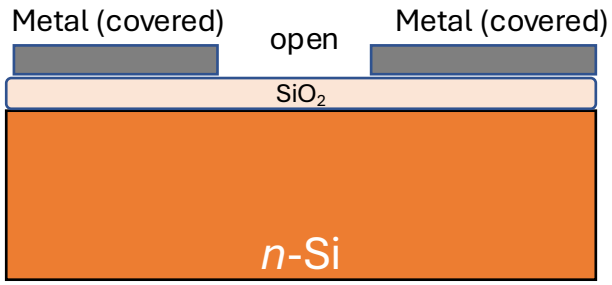


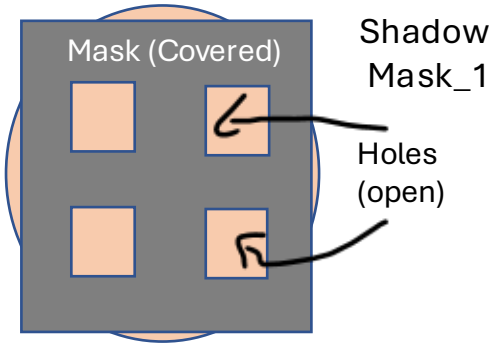


Step 3 (Litho): Open windows in SiO_2



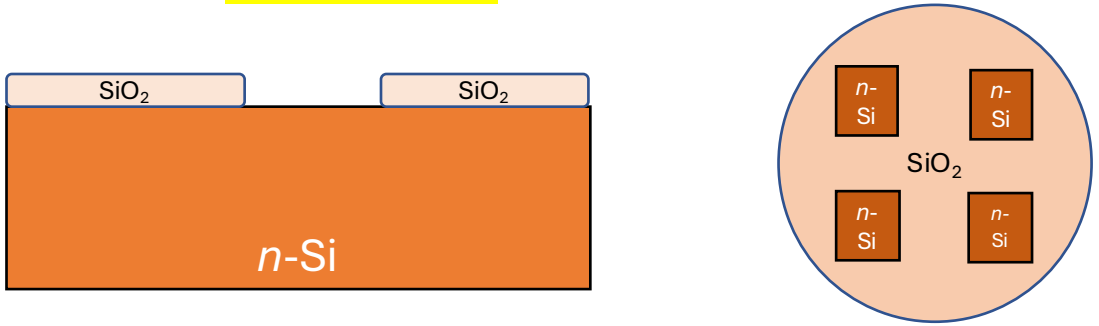
A shadow mask (metal with holes!)



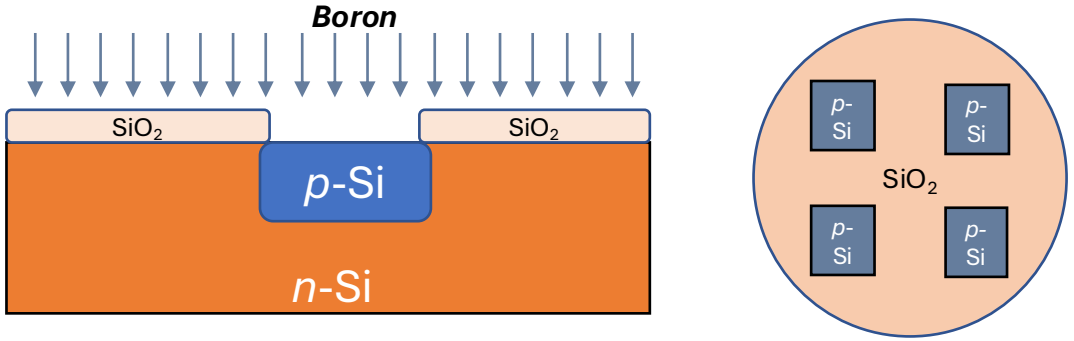


Insights from the fabrication of *pn* junction

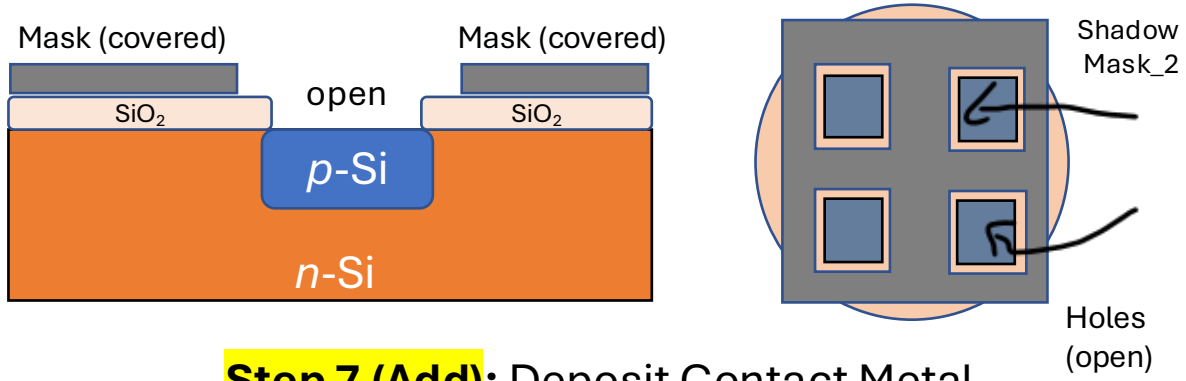
Step 4 (Sub): Remove/Etch SiO_2



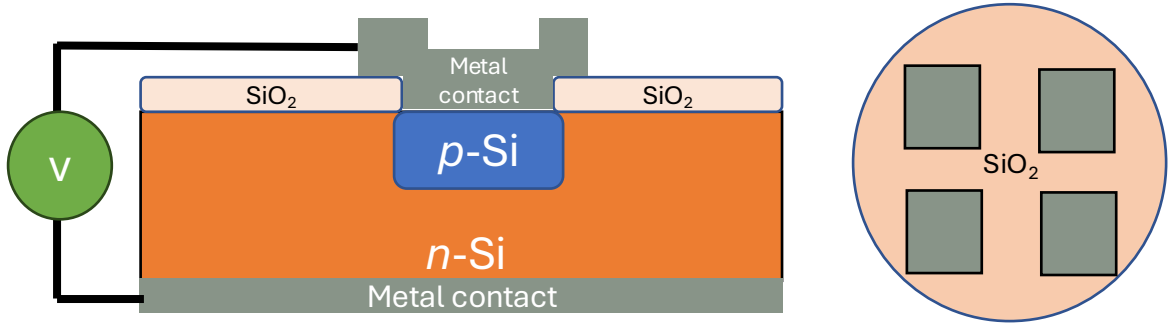
Step 5(Add): Doping open areas -> p-Si



Step 6 (Litho): Open windows for Metal deposition **(Why?)**



Step 7 (Add): Deposit Contact Metal

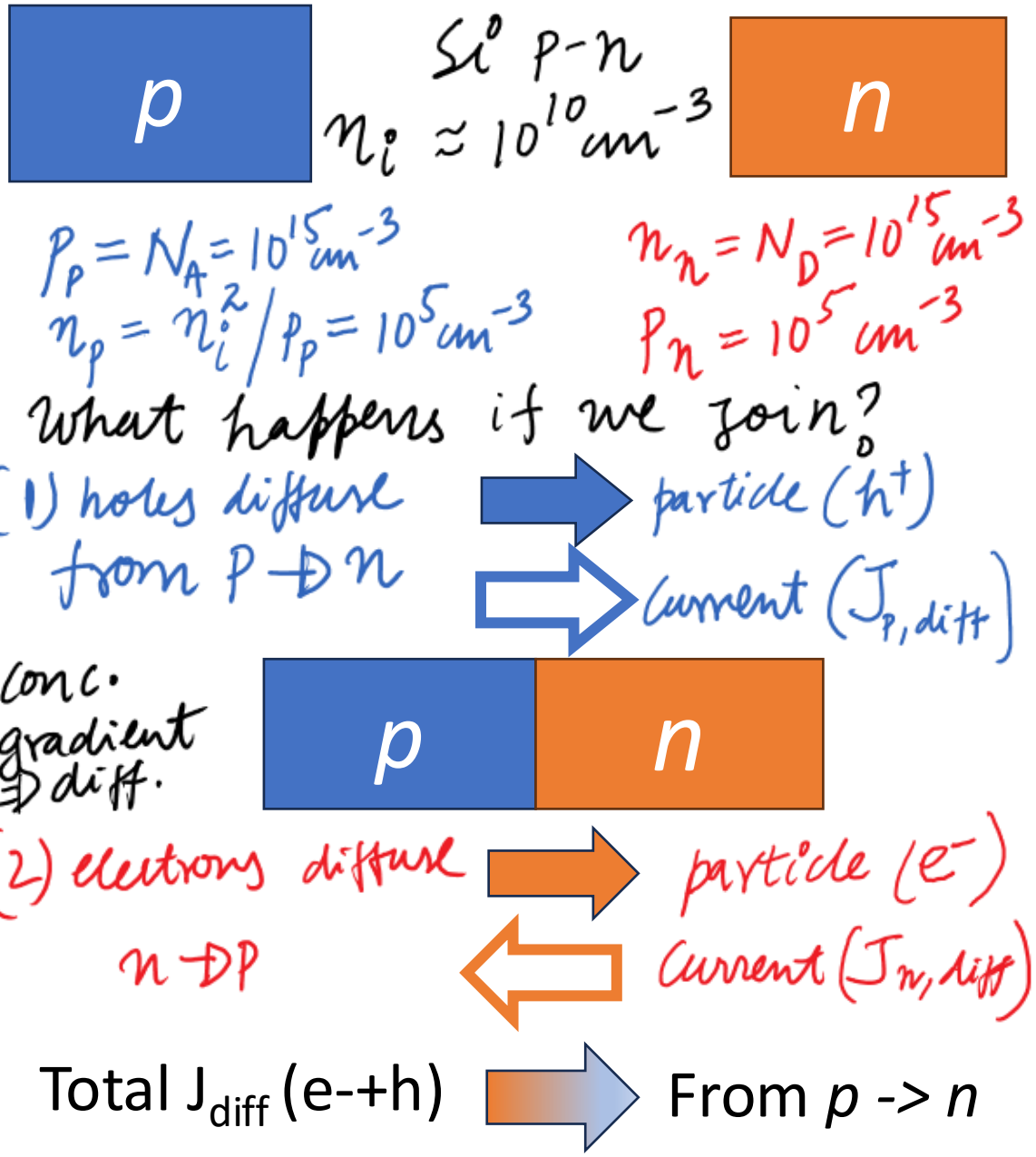


Simple Diode fabrication requires detailed planning:CMOS advanced technology node 6000 steps!!

EE2520: Intro to CMOS processing, EE5182: VLSI Technology

This Course we focus on Device Physics!

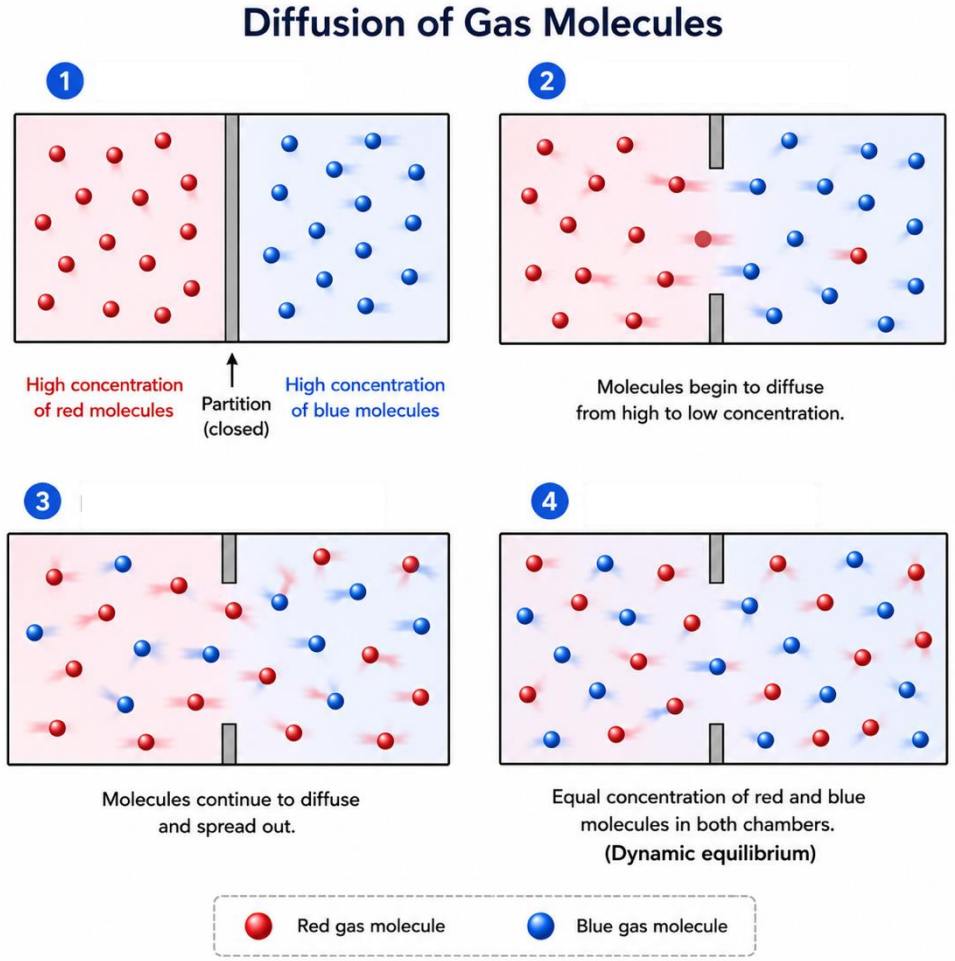
Let's start from basics: Join isolated *p* and *n*



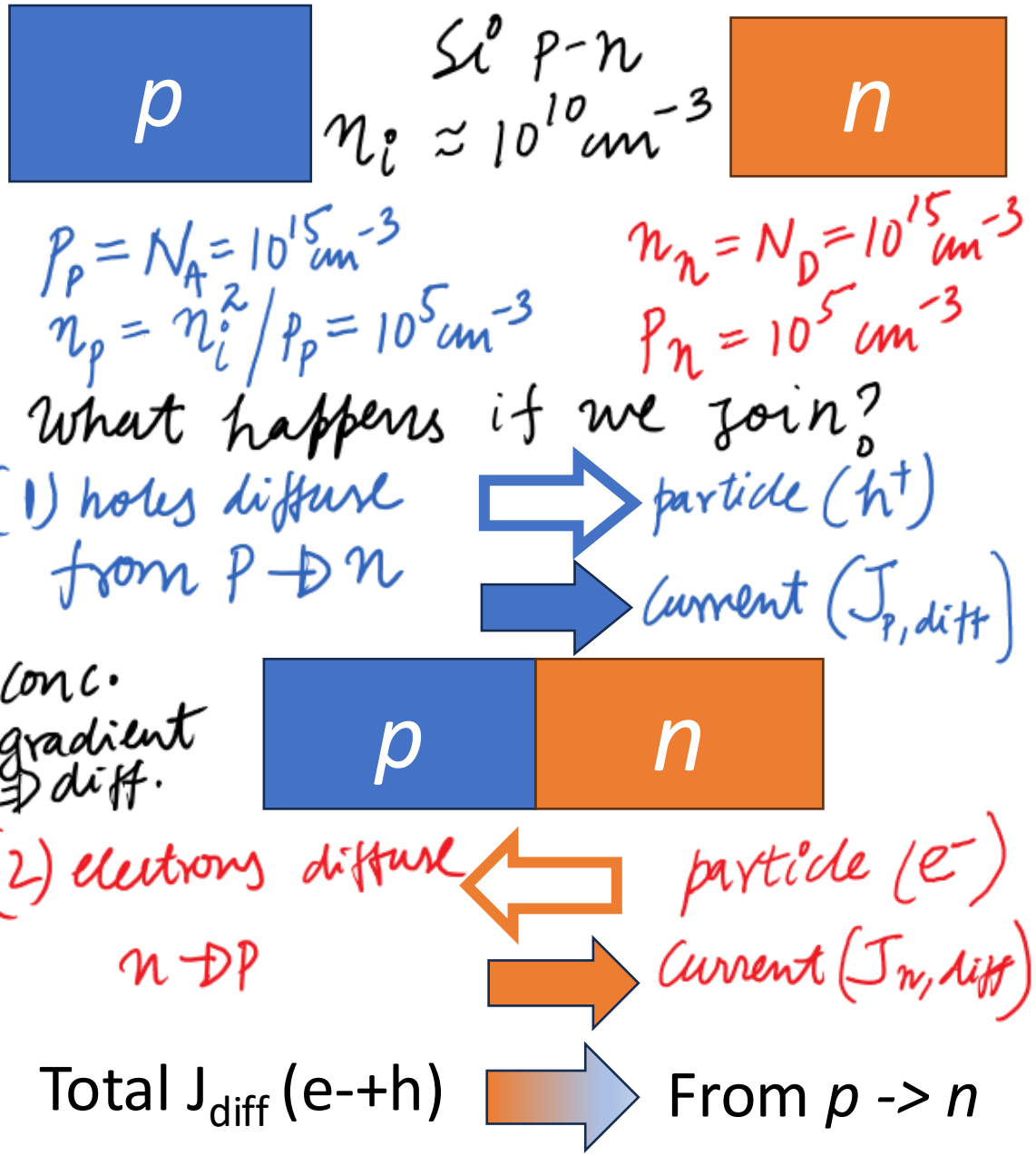
Q: Will J_{diff} continue for ever?

Common sense: No! Till grad exists

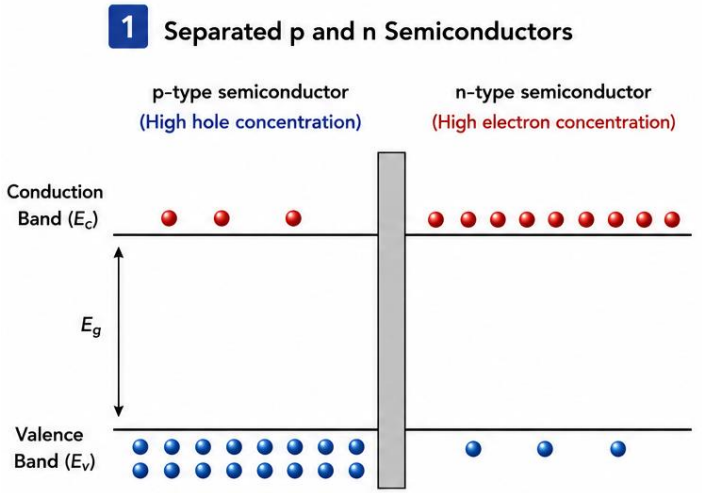
Let's do with Gas molecules



e- and h+ diffusion in joined *pn* junction



Now Let's apply for *pn* junction

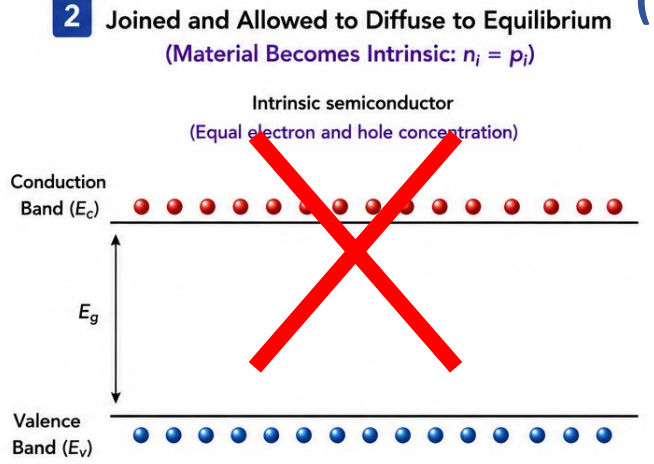


Will this scenario happen?

No! *pn*-junction are NOT useful!

Why?

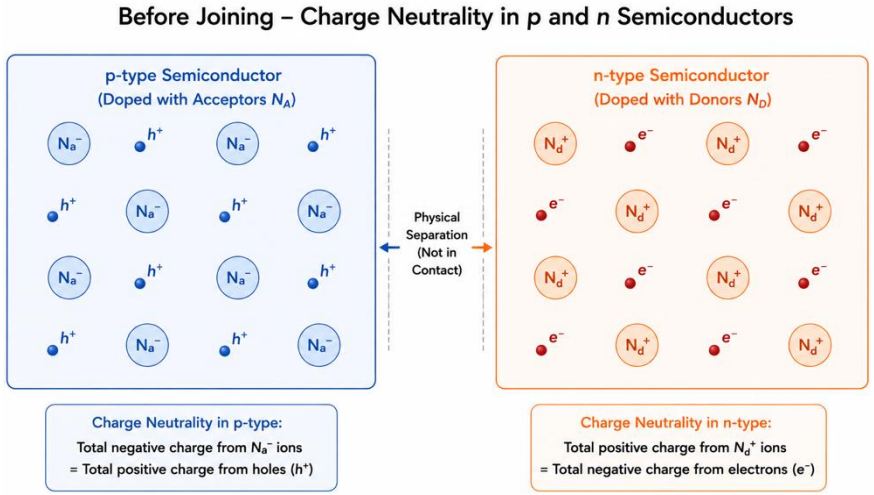
e- and h+ have charge (Gas molecules don't)

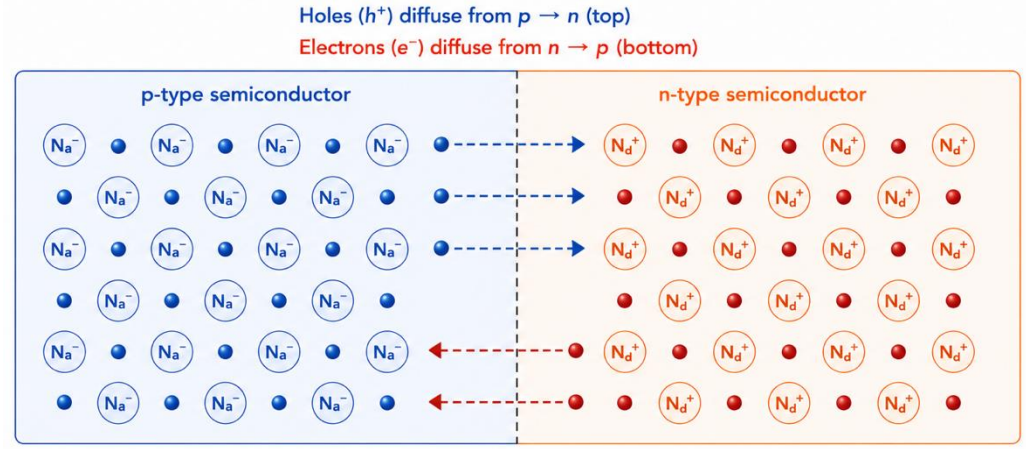


Recall:
Excess h+/e- not intrinsic to Si - comes from N_a & N_d Charge neutral!

The twist: Uncompensated Donor/Acceptor!

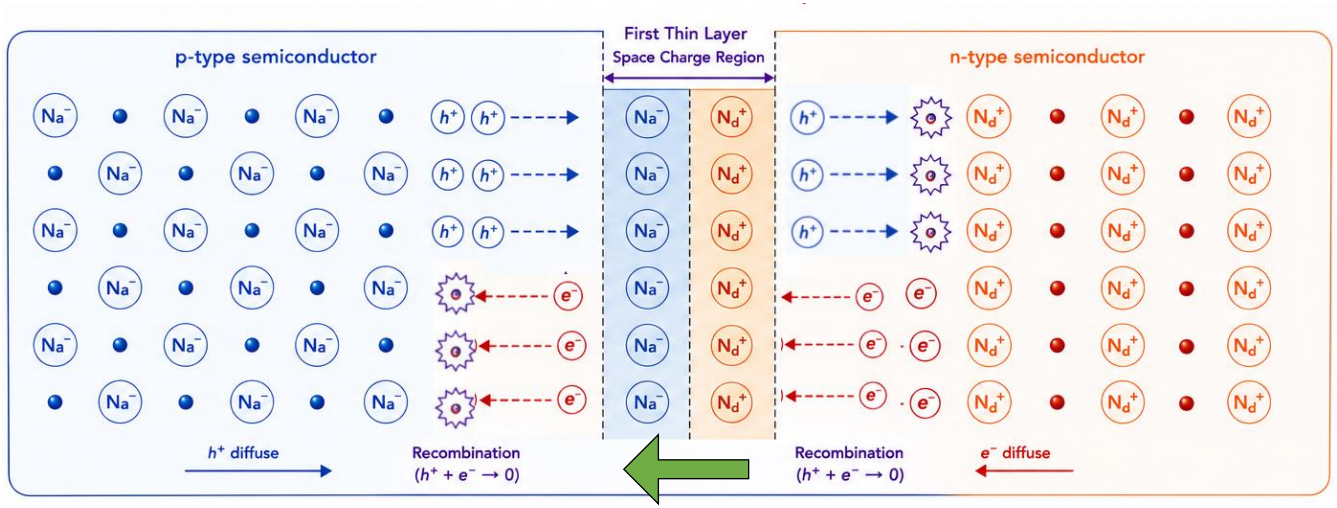
Please carefully observe charge neutrality in both p & n before contact!





Very important:
DIFFUSION of MAJORITY Carriers

Q: What happens to the diffused e- and h+?



Electric-field ξ or E

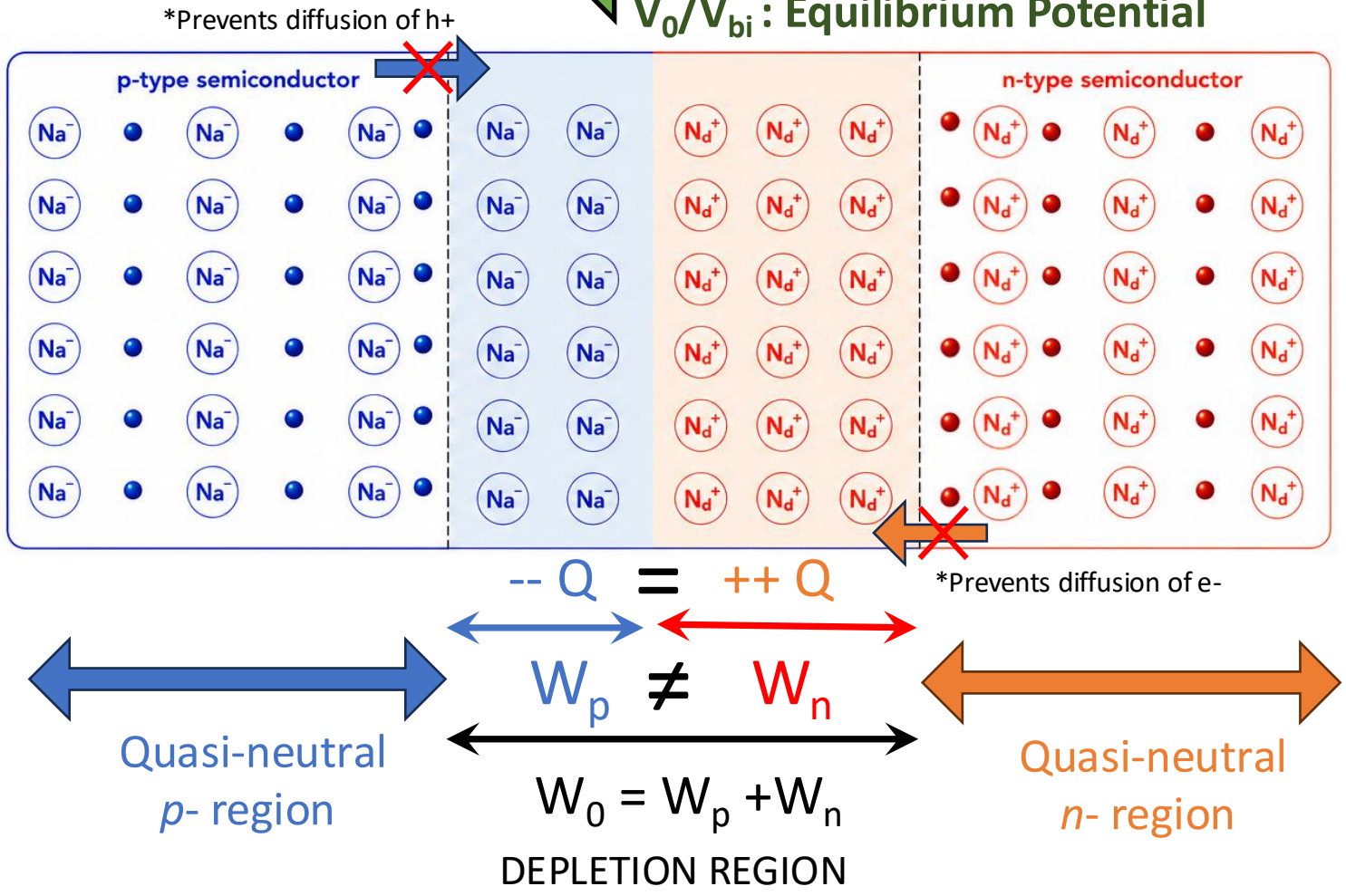
Electrostatic Potential $V = -\int E.dx$

1) h+/e- diffuses uncovers N_a-/N_d+
2) Diffused (mobile) e-,h+ recombine!
3) What remains? FIXED Uncompensated N_a- and N_d+ AT junction!
4) Consequence: E field from n -> p & Potential (V)
5) What does the E/V do? Think!
   Creates a further barrier for diffusion of e- and h+ ->
   Making the process self-limiting

Equilibrium: Diffusion vs. Drift

E_0/E_{bi} : Equilibrium Electric Field

V_0/V_{bi} : Equilibrium Potential



Insight: Diff of Maj. balanced by Drift of Min. carriers

*Prevents diffusion of h+/e- :

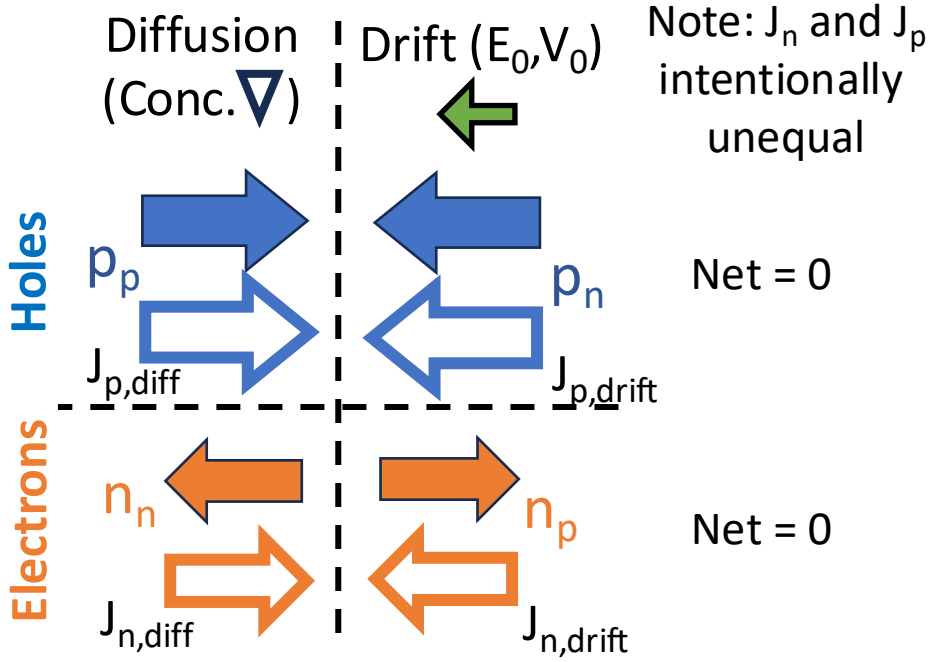
More technically correct is drift (E_0) cancel diffusion!

Diffusion: Maj (p_p (p ->n), n_n (n ->p))

What drifts? Hint: E_0,V_0 pulls/drifts n_p from p->n and p_n from n-> p.

More precisely, {particle conservation}

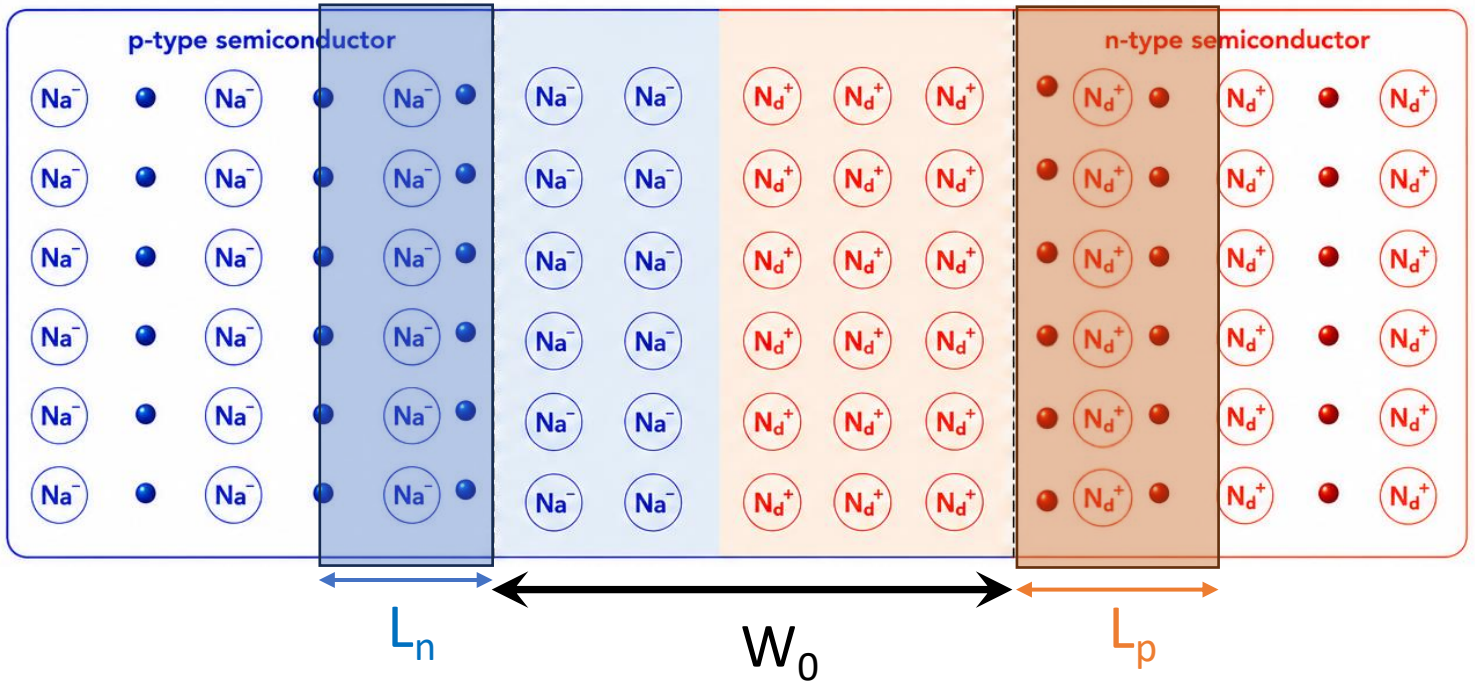
$J_{p,drift}+J_{p,diff} = 0$ & $J_{n,drift}+J_{n,diff} = 0$



Equilibrium: Built in Field/Potential blocks diffusion

E_0/E_{bi} : Equilibrium Electric Field

V_0/V_{bi} : Equilibrium Potential



Before we try to derive E_0, V_0, W_0 [N_a, N_d]
Let's ask important Recap Q1:
Q1) How 'far' diffused carriers 'travel' *before recombining**?
Q2) Where is the region from which min. carr. are "swept" from E_0 ?

** Assumption: No recombination in depletion region! (Will be revisited later)*

Answer surprisingly for both Questions:

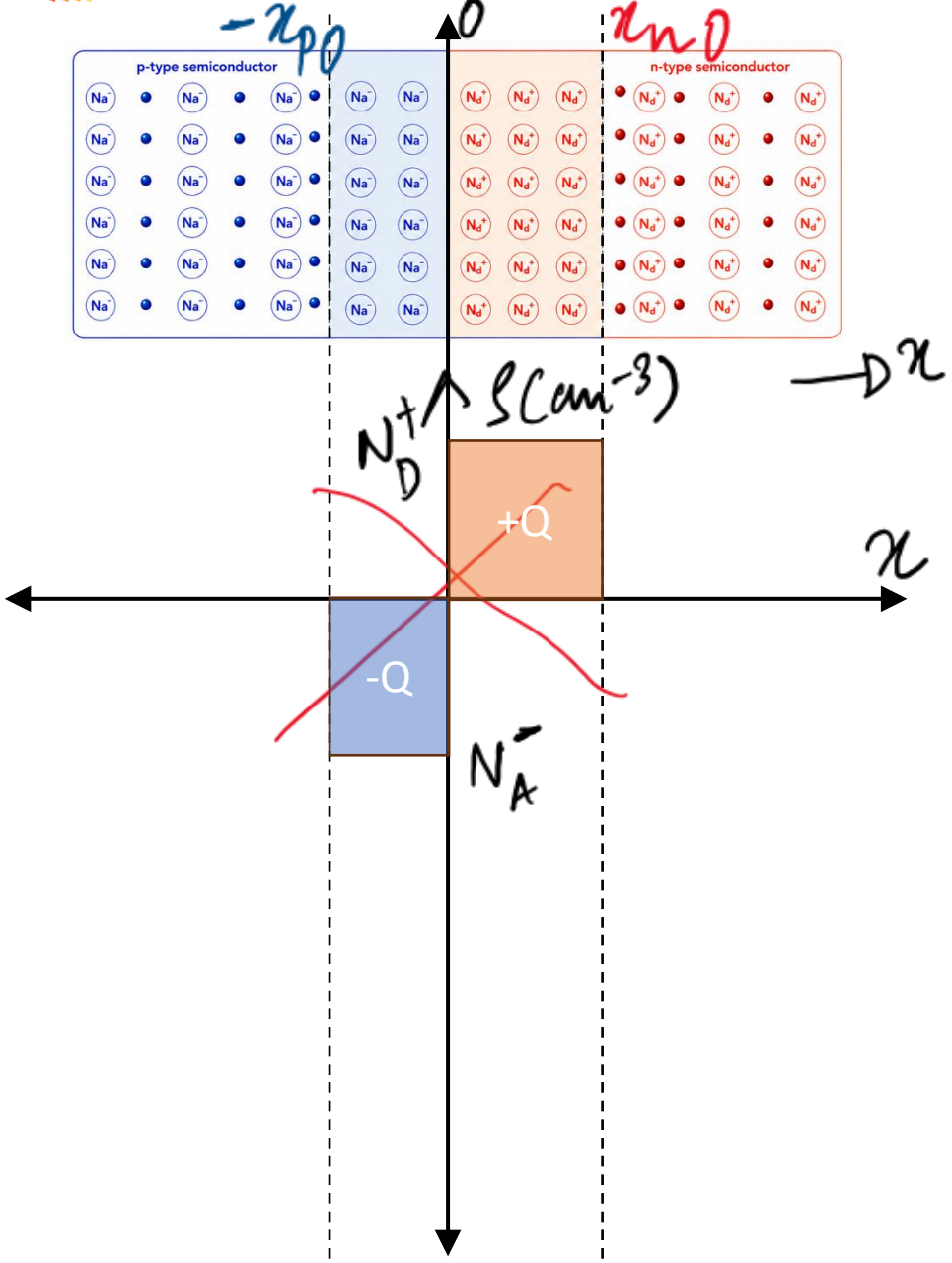
Answer :- DIFFUSSION LENGTH ($L_{p,n}$)
avg. length e^-/h^+ travels before recombining

$$L_{p,n} = \sqrt{D_{p,n}\,\tau_{p,n}}$$

How quickly carr. spread $\{cm^2 s^{-1}\}$ & Recombination life times $\{s\}$

L_p in n-side Quasi Neutral region
L_n in p-side Quasi Neutral region

Let's try to Derive V_0, E_0 and W_0 [N_a, N_d]



We will use simple electrostatics [EE1204]

1. Gauss law $\Rightarrow \nabla \cdot \bar{E} = \frac{\rho}{\epsilon_0} \overset{1D}{\Rightarrow} \frac{d\bar{E}}{dx} = \frac{\rho}{\epsilon_0}$

2. $\bar{E} = -\nabla V \overset{1D}{\Rightarrow} -\frac{dV}{dx}$ or $V = -\int \bar{E} \cdot dx$ { why -ve sign? }

So, lets plot ρ (charge density) cm^{-3}

Assuming 1) 100% of acc/donor in $-x_{p0} \Rightarrow 0$ / $0 \Rightarrow +x_{n0}$

AND 2) free carr n, p ≈ 0 in {DEPLETION REGION}

$\rho_p = N_A^- \; ; \; \rho_n = N_D^+$

Something looks not correct?

$+Q \neq -Q$ charge conservation!

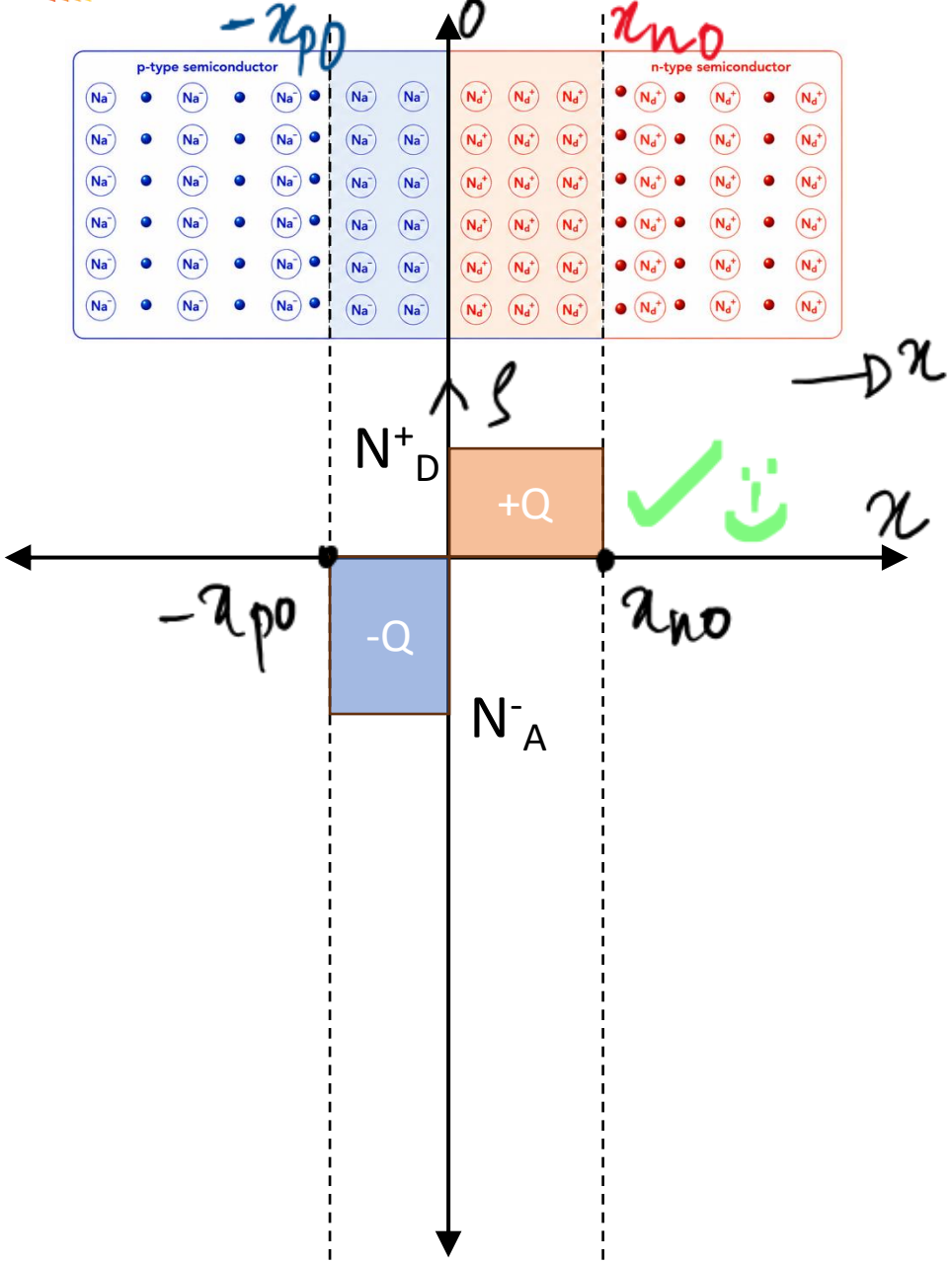
$Q_- = Q_+$

$q A x_{p0} N_A = q A x_{n0} N_D$

Since $x_n > x_p \Rightarrow N_D < N_a$

Actually $N_D < N_a$ leads to $x_n < x_p$ } larger the doping smaller the depletion width.

Let's try to Derive V_0, E_0 and W_0 [N_a, N_d]



We will use simple electrostatics [EE1204]

1. Gauss law $\Rightarrow \nabla \cdot \bar{E} = \frac{\rho}{\varepsilon_0} \overset{1D}{\Rightarrow} \frac{d\bar{E}}{dx} = \frac{\rho}{\varepsilon_0}$

2. $\bar{E} = -\frac{dV}{dx}$ or $V = -\int \bar{E} \cdot dx$ { why -ve sign? }

$$Q_- = Q_+$$

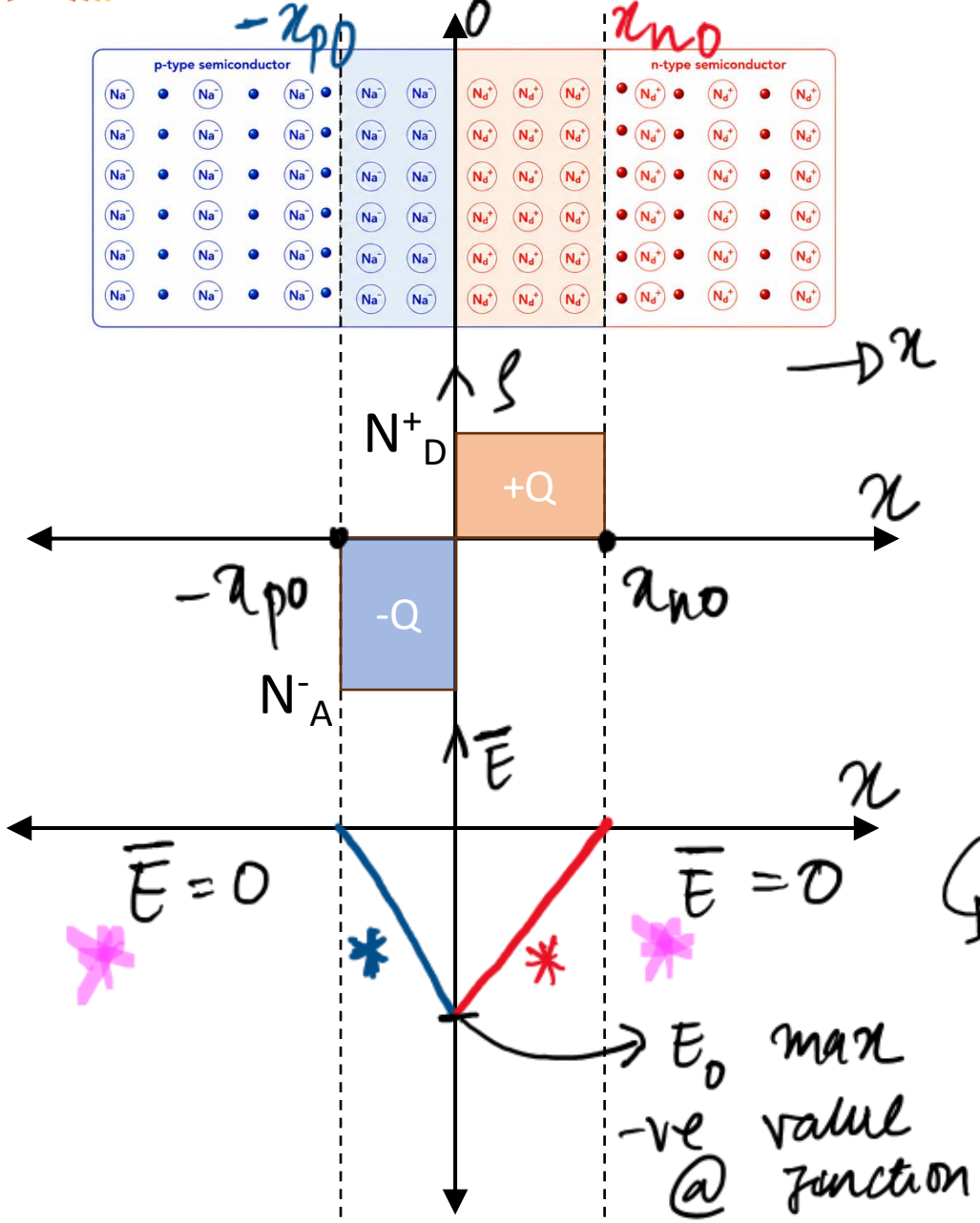
$$q A x_{p0} N_A = q A x_{n0} N_D$$

In addition to,

$$x_{p0} = \frac{x_{n0} N_D}{N_A}; \qquad x_{p0} + x_{n0} = W_0$$

$$x_{n0} = W_0\left(\frac{N_A}{N_A + N_D}\right) \text{ \& } x_{p0} = W_0\left(\frac{N_D}{N_A + N_D}\right)$$

Let's try to Derive V_0, E_0 and W_0 [N_a, N_d]



1) $\frac{d\bar{E}}{dx} = \frac{\rho}{\epsilon_0}$ 2) $V = -\int \bar{E}\cdot dx$

Since we know $\rho \Rightarrow \bar{E}$ profile

Recall if parallel plate with $+\sigma$, $-\sigma$
$E(x) \Rightarrow$ constant $\sigma/\epsilon_0 \epsilon$; $V(x) \propto x$ $\Delta V = \frac{\sigma d}{\epsilon_0}$

Here $0 - N_A \rightarrow -x_{p0} < x < 0$

$\rho(x) \Rightarrow +N_D \rightarrow 0 < x < x_{n0}$

$\Rightarrow 0 \rightarrow$ B.N regions $x < -x_{p0}$; $x > x_{n0}$

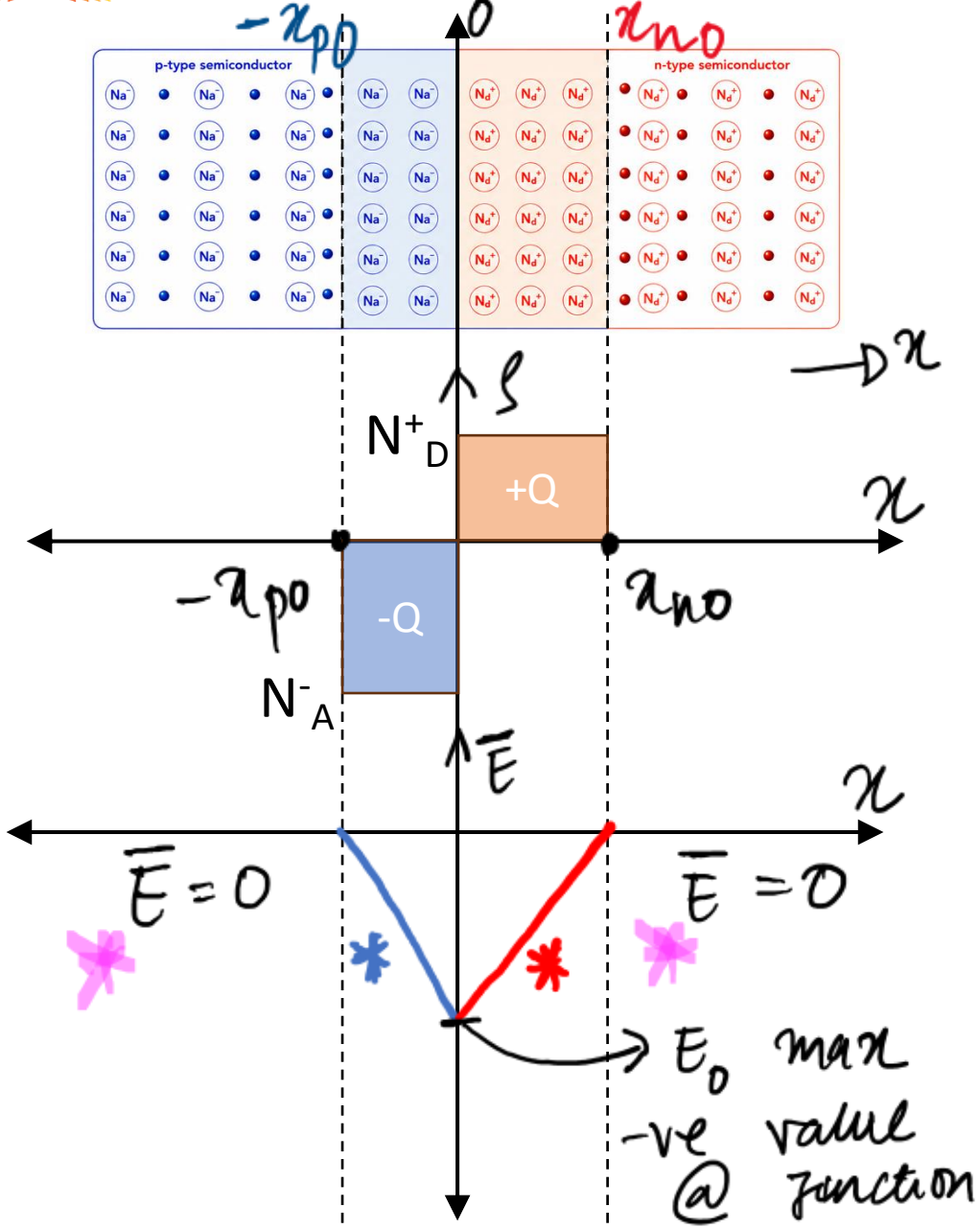
Imp: gives us limits of integration.

$\hookrightarrow * \frac{d\bar{E}}{dx} = -\frac{q}{\epsilon} N_A$ $\quad -x_{p0} < x < 0 \rightarrow \bar{E}$ fixed -ve slope

$* \frac{d\bar{E}}{dx} = +\frac{q}{\epsilon} N_D$ $\quad 0 < x < x_{n0} \Rightarrow \bar{E}$ fixed +ve slope

What does is it physically mean?

Let's try to Derive V_0, E_0 and W_0 [N_a, N_d]



1) $\frac{d\bar{E}}{dx} = \frac{\rho}{\epsilon_0}$ 2) $V = -\int \bar{E}\cdot dx$

* $\frac{d\bar{E}}{dx} = -\frac{q}{\epsilon} N_A$ $\quad -x_{p0} < x < 0 \Rightarrow \bar{E}$ fixed -ve slope

* $\frac{d\bar{E}}{dx} = +\frac{q}{\epsilon} N_D$ $\quad 0 < x < x_{n0} \Rightarrow \bar{E}$ fixed +ve slope

Value of $\bar{E}_{0,max}$? Integration above diff equations.

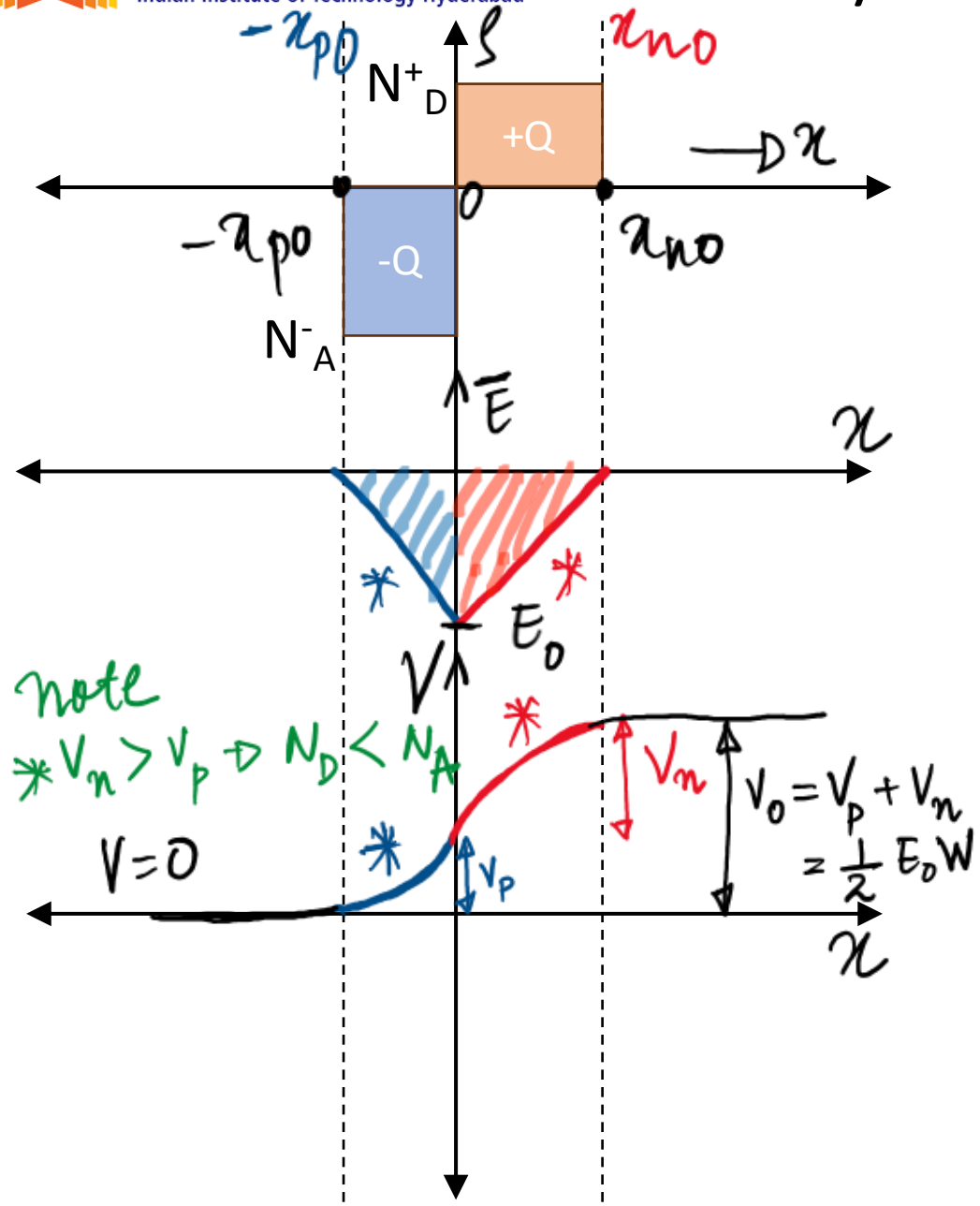
$$\int_0^{E_0} d\bar{E} = -\frac{q}{\epsilon} N_A \int_{-x_{p0}}^{0} dx \qquad \int_{E_0}^{0} d\bar{E} = +\frac{qN_D}{\epsilon} \int_0^{+x_{n0}} dx$$

$-x_{p0} < x < 0$ $\qquad 0 < x < +x_{n0}$

$$E_{0\,max\,@\,junc} = -\frac{q}{\epsilon} N_A x_{p0} = \frac{-q}{\epsilon} N_D x_{n0} = \frac{-q}{\epsilon} \frac{N_A N_D}{N_A + N_D} W_0$$

Self study: Question — why did we assume same E_0 or why E_0 is continuous?

Let's try to Derive V_0, E_0 and W_0 [N_a, N_d]



Next, let's focus on V_0 $\quad V = -\int \bar{E}.dx$

$$V_p = \int_{-x_{p0}}^{0} \bar{E}.dx = \frac{1}{2}\frac{q}{\epsilon} N_A x_{p0}^2 = \frac{1}{2} E_0 x_{p0} \; *$$

$$V_n = \int_{0}^{+x_{n0}} \bar{E}.dx = \frac{1}{2}\frac{q}{\epsilon} N_D x_{n0}^2 = \frac{1}{2} E_0 x_{n0} \; *$$

*Imp Insight $V_{p,n} \propto x^2_{p0,n0} \propto \frac{1}{N_{A,D}}$

Higher the doping, lower the V drop.

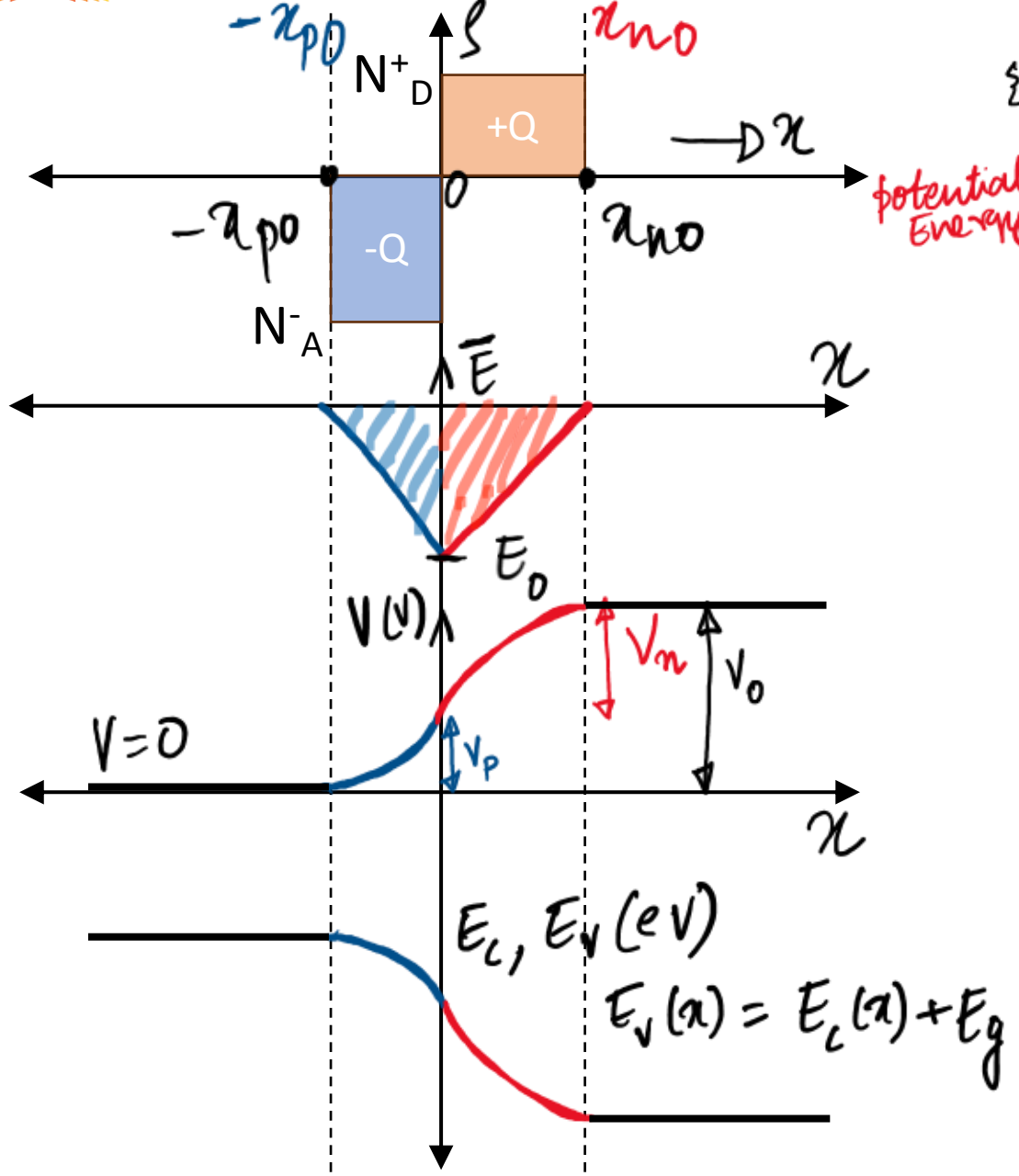
$$V_0 = V_p + V_n = \frac{1}{2}\frac{q}{\epsilon}\left[N_A x_{p0}^2 + N_D x_{n0}^2\right]$$

$$V_0 = \frac{1}{2} E_0 (x_{n0} + x_{p0}) = \frac{1}{2} E_0 W_0 \left(\begin{matrix}\text{red + blue}\\ \text{areas}\end{matrix}\right)$$

Substituting $E_0(W_0)$

$$V_0 = \frac{1}{2}\frac{q}{\epsilon}\frac{N_A N_D}{N_A + N_D} W_0^2$$

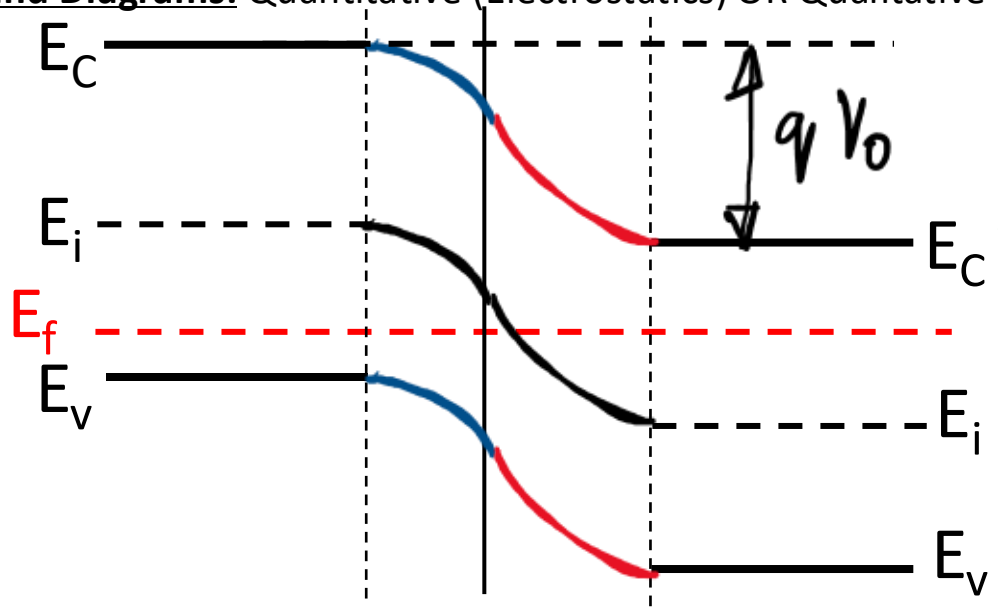
Equil. Energy Band Diagram (4 steps)



E_c, E_v ⇒ Electrostatic potential
{Energy bands} Energy for "electrons" (in eV) ↗ later!

potential Energy → $E = (-q)\,V$ ← potential ; ∴ $E_c(x) = -qV(x) + \text{const.}$
charge
[Historical Baggage!]

Band Diagrams: Quantitative (Electrostatics) OR Qualitative (Intuition)



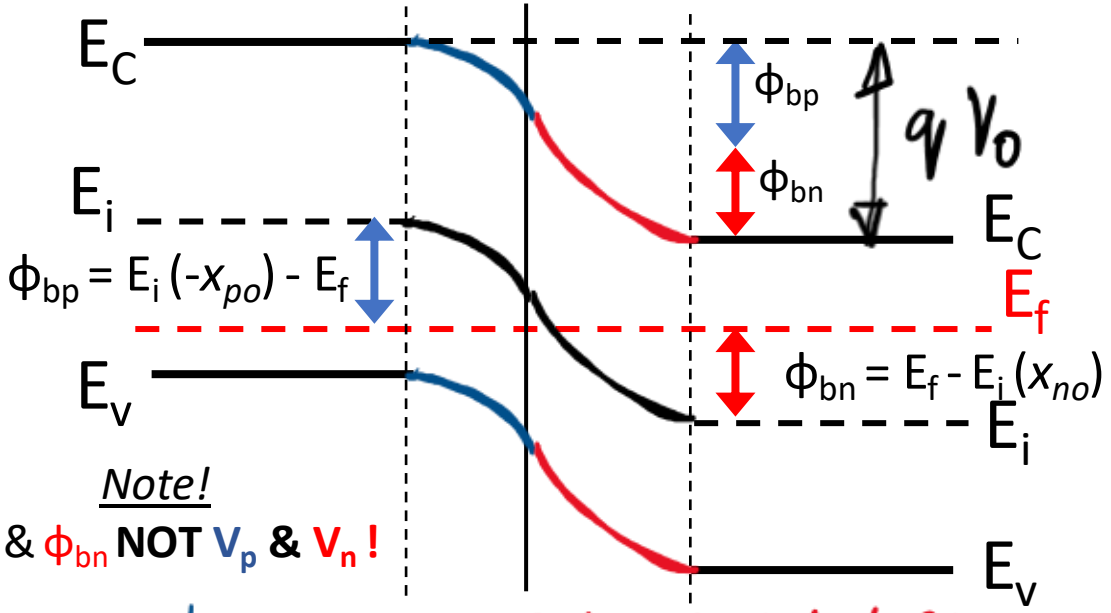
V. Imp to develop this intuition!
See: Next Slide!

STEP 1: E_f - no gradient (Equilibrium)
STEP 2: Draw Flat $E_{c,v,i}$ in Q.N reg.
STEP 3: Location of physical junc. –
Depletion extends farther into the lightly doped side
STEP 4: Band bending in depletion reg.

Built in Potential $V_0(N_A,N_D)$

Notice: V_0 and E_0 function of W_0; W_0 not directly from N_A,N_D – No closed form solution

A. Easiest to deduce $V_0(N_A,N_D)$ - Band Diagram!



$qV_0 = \phi_{Bp} + \phi_{Bn} \Rightarrow$ Body potential of P+n $\Rightarrow$ Total band bending (V_0)

$$V_0 = \frac{k_B T}{q}\left[\ln\left(\frac{p_p = N_A}{n_i}\right) + \ln\left(\frac{n_n = N_D}{n_i}\right)\right]$$

$$V_0 = \frac{k_B T}{q}\ln\left[\frac{N_A N_D}{n_i^2}\right]$$

B. Alternate derivation $V_0(N_A,N_D) \rightarrow J_{drift} = J_{diff}$

Recall, $J_{p,drift} = J_{p\,diff}$ & $J_{n,drift} = J_{n,\,diff}$.

$$q\mu_p p(x) E(x) = qD_p \frac{dp(x)}{dx}$$

Subs $E(x) = -\frac{dV(x)}{dx}$ & $D_p/\mu_p = \frac{k_B T}{q}$

$$-\frac{q}{k_B T}\int_0^{V_0} dV = \int_{p_p}^{p_n} \frac{1}{p}\,dp$$

{ Note limits of $\int -x_{p0} \rightarrow x_{n0}$ }

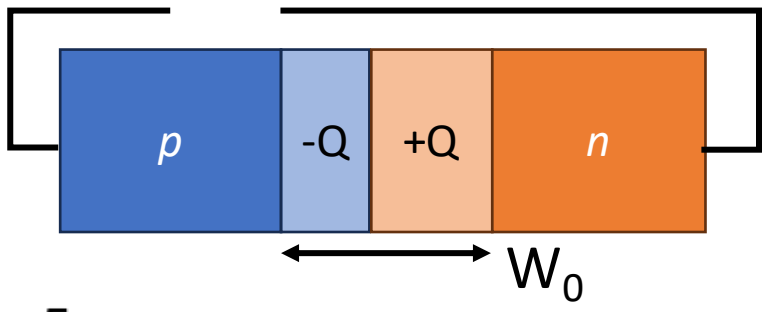
$$V_0 = \frac{k_B T}{q}\ln\left(\frac{p_p}{p_n}\right) = \frac{k_B T}{q}\ln\left(\frac{N_A N_D}{n_i^2}\right)$$

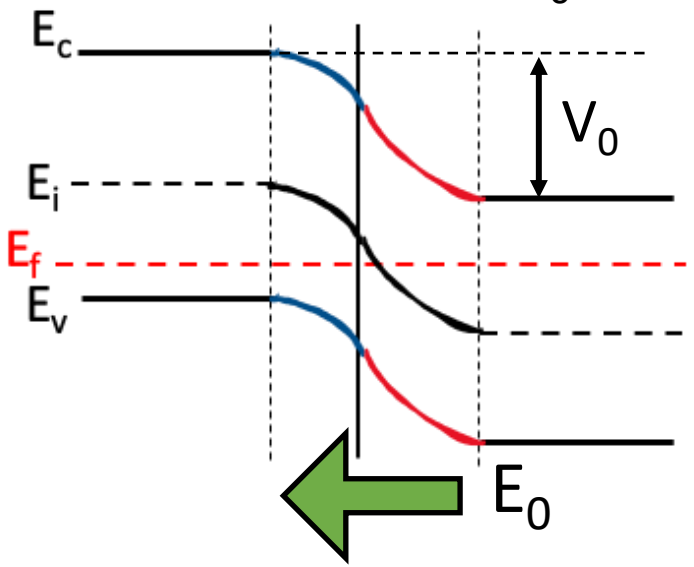
$$W_0 = \sqrt{\frac{2\epsilon k_B T}{q^2}\left(\frac{N_A + N_D}{N_A N_D}\right)\ln\frac{N_A N_D}{n_i^2}}\;;\; E_0 = \frac{2V_0}{W_0}$$

[Do we need to memorise ??]

Non-Equilibrium Effect of V_{bias} – on E,V,W

(a) Equilibrium: $V_{bias} = 0$

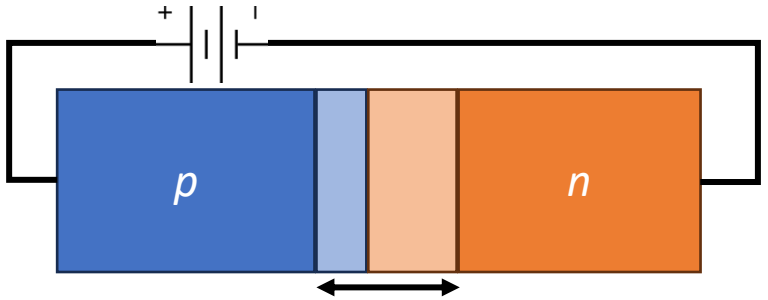




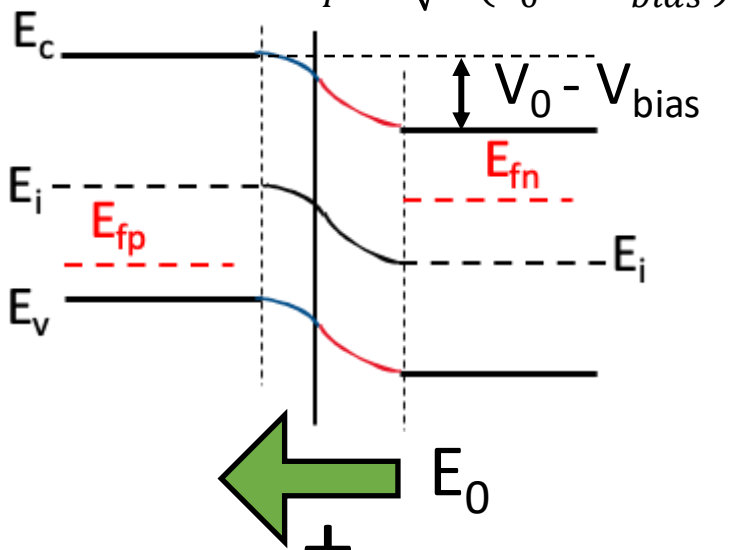
$$V_0 = V_t \ln(N_A N_D / n_i^2)$$

$$W_0 = \sqrt{\alpha V_0}\ ;\ \alpha = \frac{2\epsilon}{q}\left(\frac{1}{N_A} + \frac{1}{N_D}\right)$$

(b) FWD: $V_{bias} > 0$



$$W_F = \sqrt{\alpha(V_0 - V_{bias})}$$



Split in Fermi Levels

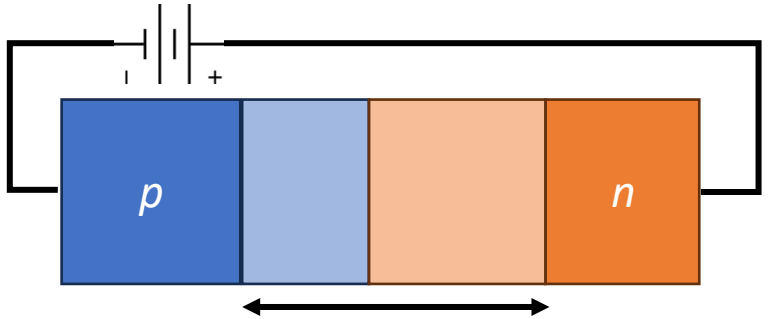
$E_{fn} - E_{fp} = qV_{bias}$

OR

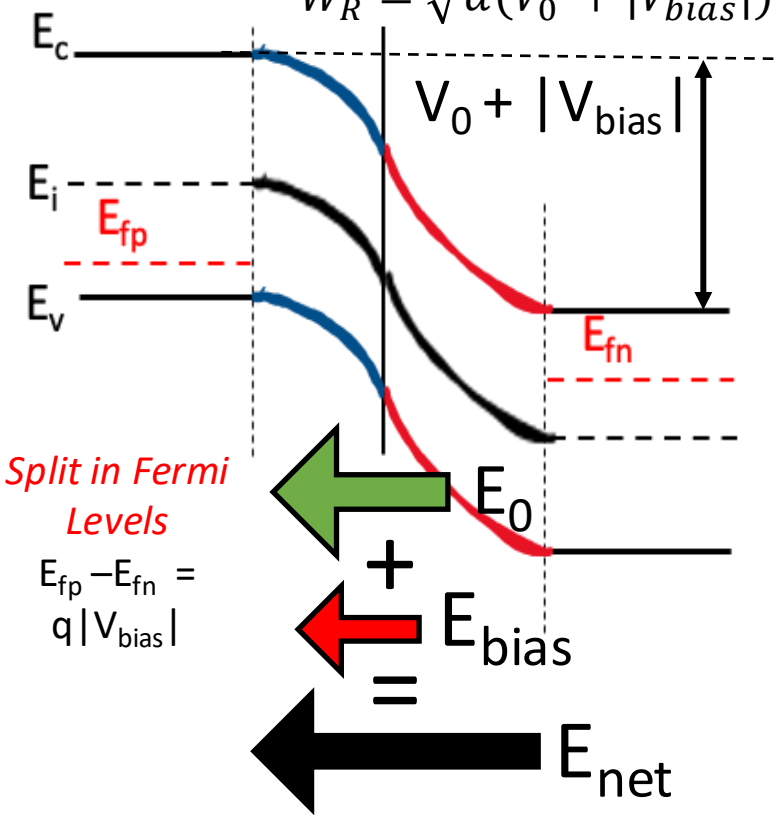
$E_{fn} - E_{fp} = V_{bias}$

Depends on UNITs

(c) REV $V_{bias} < 0$



$$W_R = \sqrt{\alpha(V_0 + |V_{bias}|)}$$



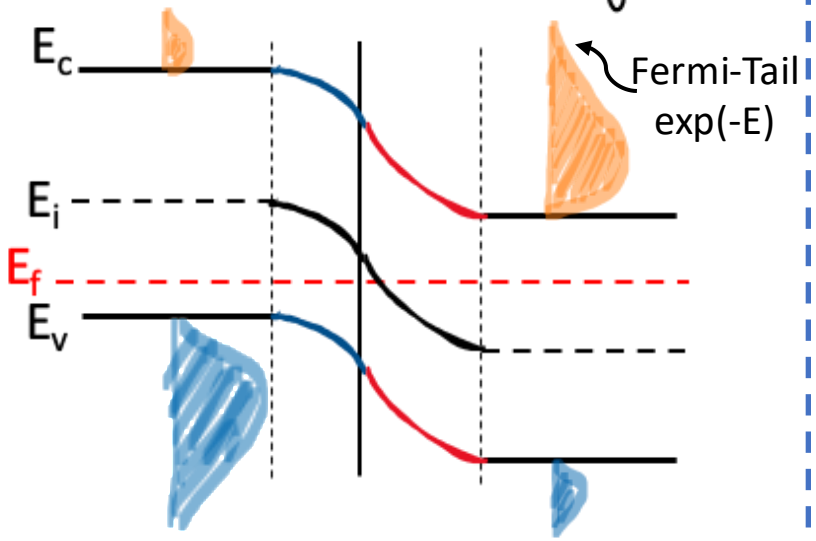
Split in Fermi Levels

$E_{fp} - E_{fn} = q|V_{bias}|$

Non-Equilibrium I(V_{bias}) – Qualitative

(a) Equilibrium: V_{bias} = 0

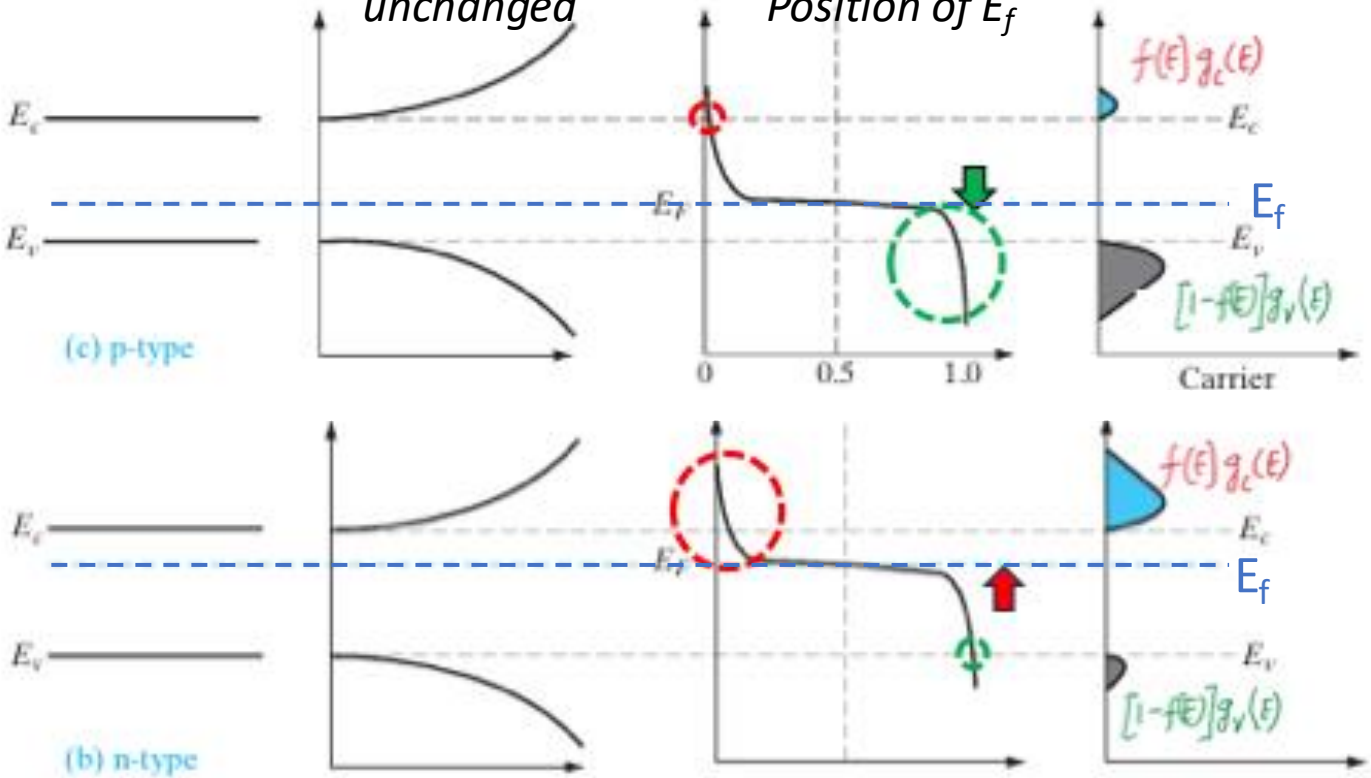
We re-introduce $n(E) = g(E)f(E)$



Recap: n(E) = g(E)f(E) – free carr. avl for conduction per energy level

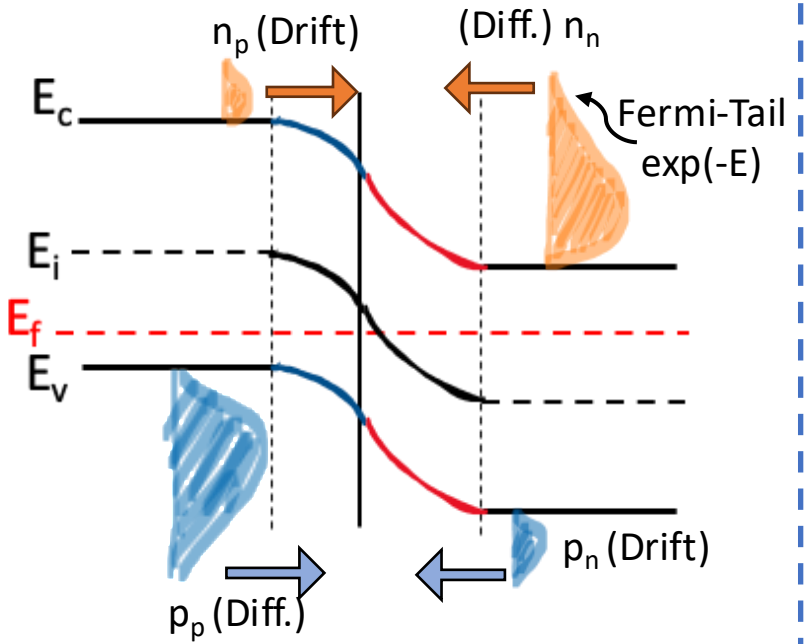
g(E) ∝ $E^{1/2}$ *unchanged*

f(E) ∝ $e^{-(E-Ef)/kT}$ *Position of E_f*



Non-Equilibrium I(V_{bias}) – Qualitative

(a) Equilibrium: $V_{bias} = 0$



Jp_p (Diff.) = Jp_n(Drift)

Jn_n(Diff.) = Jn_p (Drift)

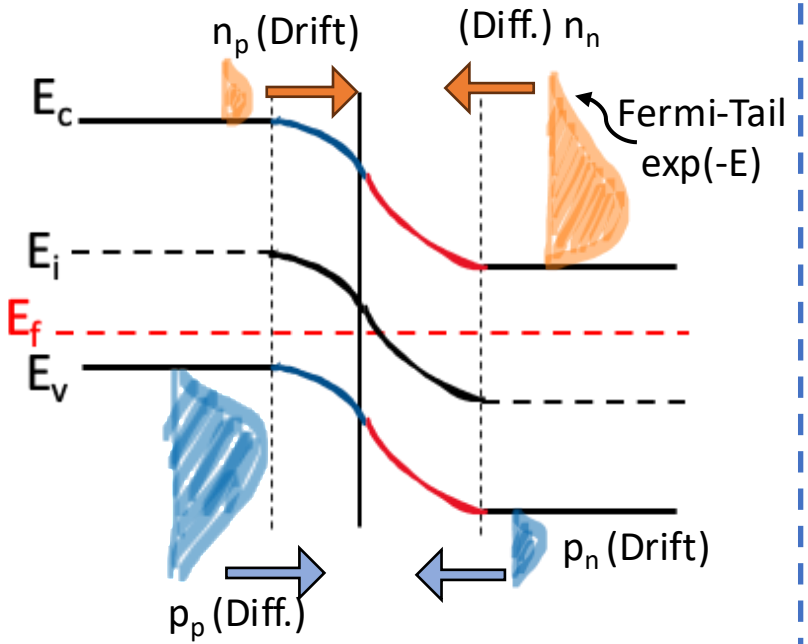
$I_{net} = 0$

(b) FWD: $V_{bias} > 0$

(c) REV $V_{bias} < 0$

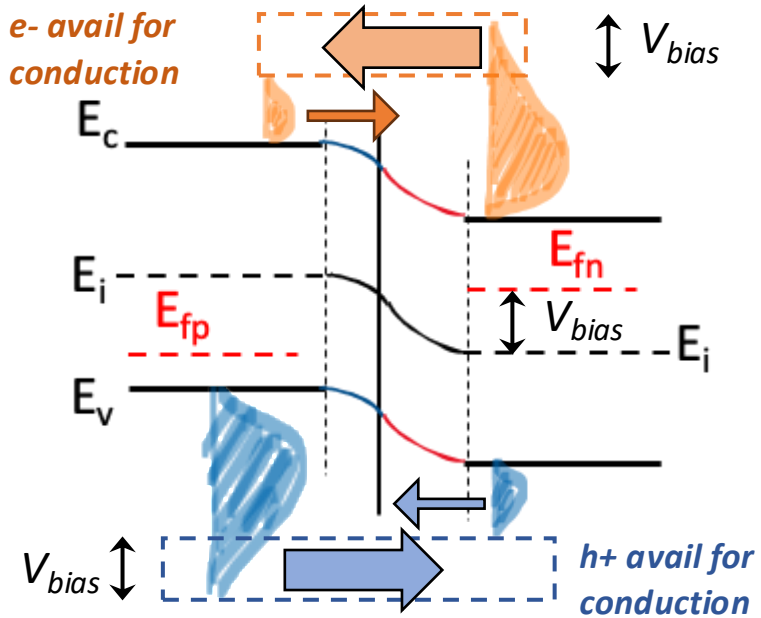
Non-Equilibrium I(V_{bias}) – Qualitative

(a) Equilibrium: $V_{bias} = 0$



Jp_p (Diff.) = Jp_n(Drift)
Jn_n(Diff.) = Jn_p (Drift)
$I_{net} = 0$

(b) FWD: $V_{bias} > 0$



p_p and $n_n \propto \exp(qV_{bias})$ – (Fermi-tail)
Uncovered/available for Diff.
p_n and n_p *Drift unchanged* – (Why?)*

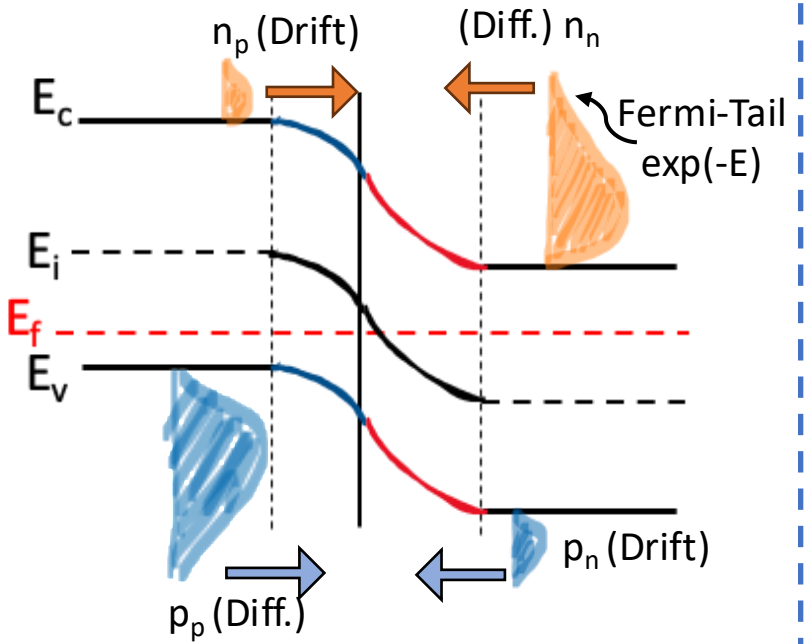
Jp_p (Diff.) $\propto exp(qV_{bias}) >> Jp_n$(Drift)
Jn_n(Diff.) $\propto exp(qV_{bias}) >> Jn_p$ (Drift)

$I_{FWD} \propto \exp(qV_{bias})$: Diff., Maj. Carr

(c) REV $V_{bias} < 0$

Non-Equilibrium I(V_{bias}) – Qualitative

(a) Equilibrium: $V_{bias} = 0$

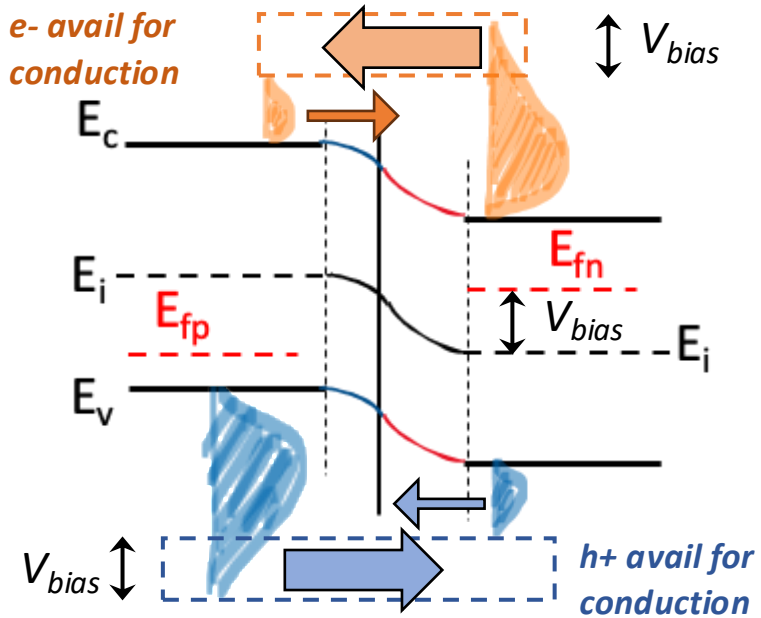


Jp_p (Diff.) = Jp_n(Drift)
Jn_n(Diff.) = Jn_p (Drift)
$I_{net} = 0$

	Expected	In *pn* Junc
J_{drift}	$\propto$ E, V_{bias},	Indep
J_{diff}	Indep E, V_{bias}	$\propto \exp(V_{bias})$



(b) FWD: $V_{bias} > 0$

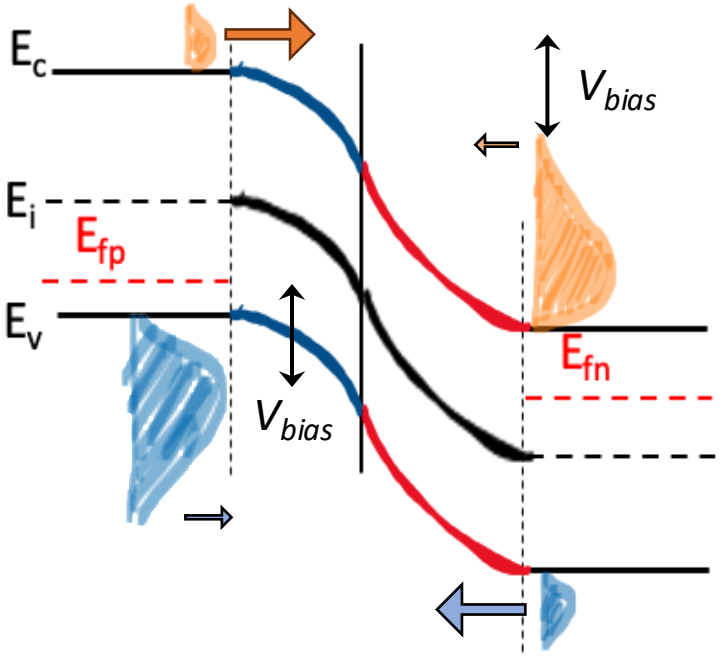


p_p and $n_n \propto \exp(qV_{bias})$ – (Fermi-tail)
Uncovered/available for Diff.
p_n and n_p *Drift unchanged* – (Why?)*

Jp_p (Diff.) $\propto exp(qV_{bias}) >> Jp_n$(Drift)
Jn_n(Diff.) $\propto exp(qV_{bias}) >> Jn_p$ (Drift)

$I_{FWD} \propto \exp(qV_{bias})$: Diff., Maj. Carr

(c) REV $V_{bias} < 0$



p_p and $n_n \propto \exp(qV_{bias})$ – (Fermi-tail)
Uncovered/available for Diff.
p_n and n_p *Drift unchanged* – (Why?)*

Jp_p (Diff.) $\propto exp(-qV_{bias}) << Jp_n$(Drift)
Jn_n(Diff.) $\propto exp(-qV_{bias}) << Jn_p$ (Drift)

$I_{REV} = I_0$: Drift., Min. Carr

J_{drift} J_{diff} vs. V_{bias}

Example: Volunteers '*pulling*' donations



Scenario 1:
Donations independent of #Volunteers



Scenario 2:
Donations increase with #Volunteers
What is happening?

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भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

Why is J_{drift} independent of E_{net}/V_{bias}

Example: Volunteers '*pulling*' donations

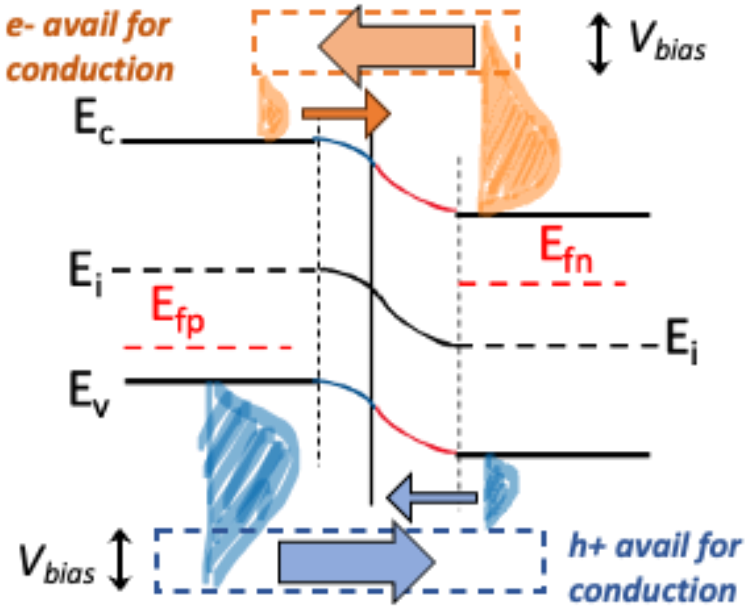


Classic supply side problem!

Scenario 1:
Donations independent of #Volunteers
Sunday Morning: Less people on street!



Scenario 2:
Donations increase with #Volunteers
Weekday Morning: Large people on street!



$J_{drift} \propto E/V_{bias}$ assumes plenty of carr!

Not true: p_n,n_p generated within L_p, L_n available to be swept by E.

Very limited, J_{drift} const E.
$I_{REV} = I_0$: Drift., Min. Carr

More technically,
$\tau_{gen} >> \tau_{transit}$ for even small E

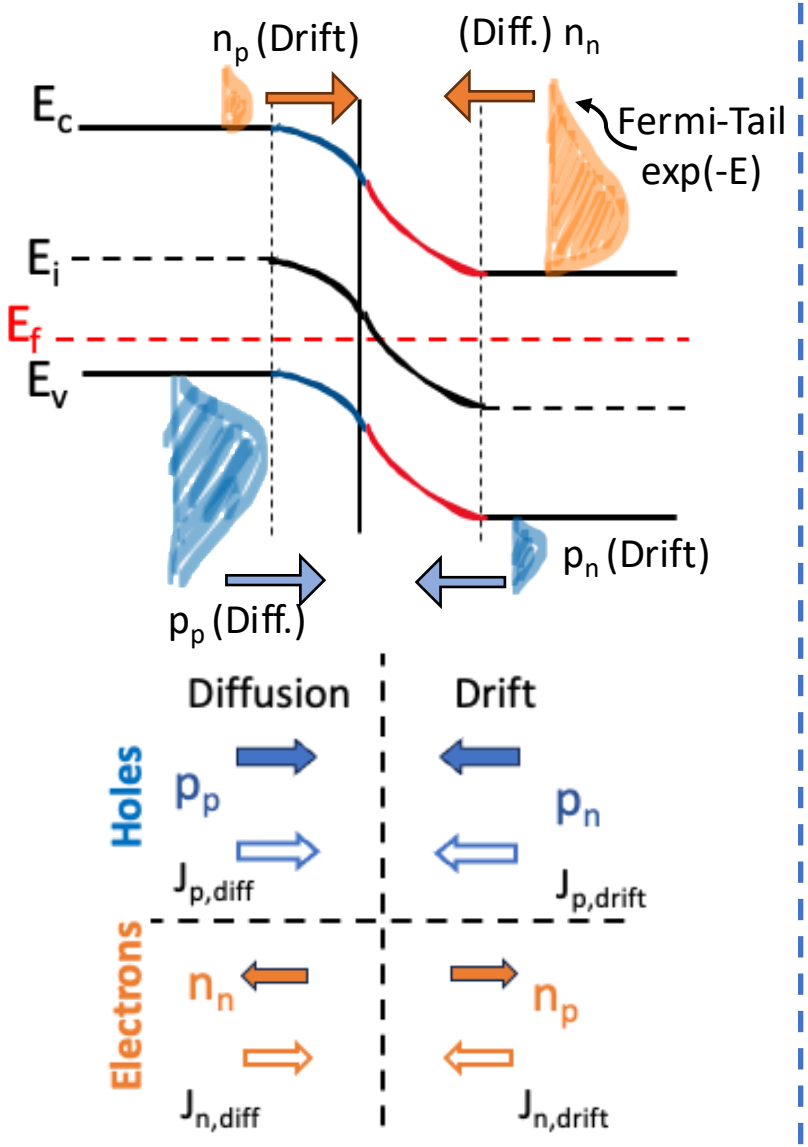
J_{diff} independent of V_{bias} assumes no barrier to diffusion

Not true: V_{bias} modulates barrier which exponentially (un)covers n_n,p_p Maj. Carr.

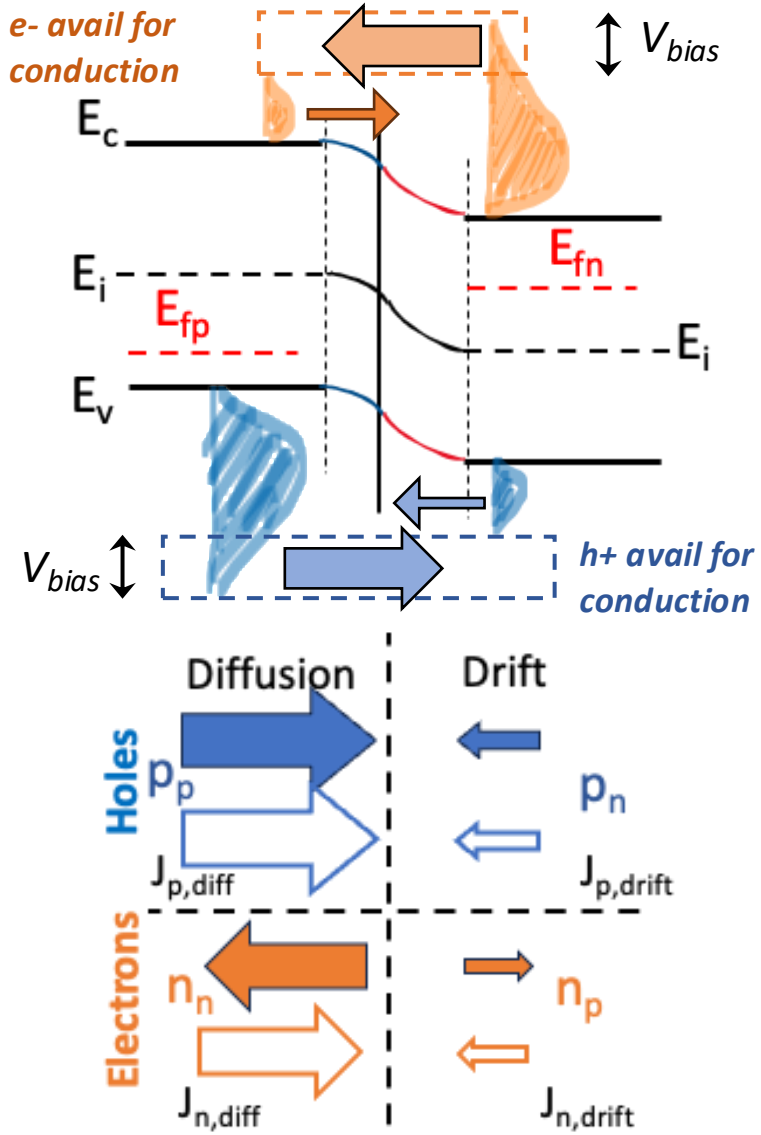
Therefore,
$I_{FWD} \propto \exp(qV_{bias})$: Diff., Maj. Carr

Non-Equilibrium – e-/h+ & Current Components

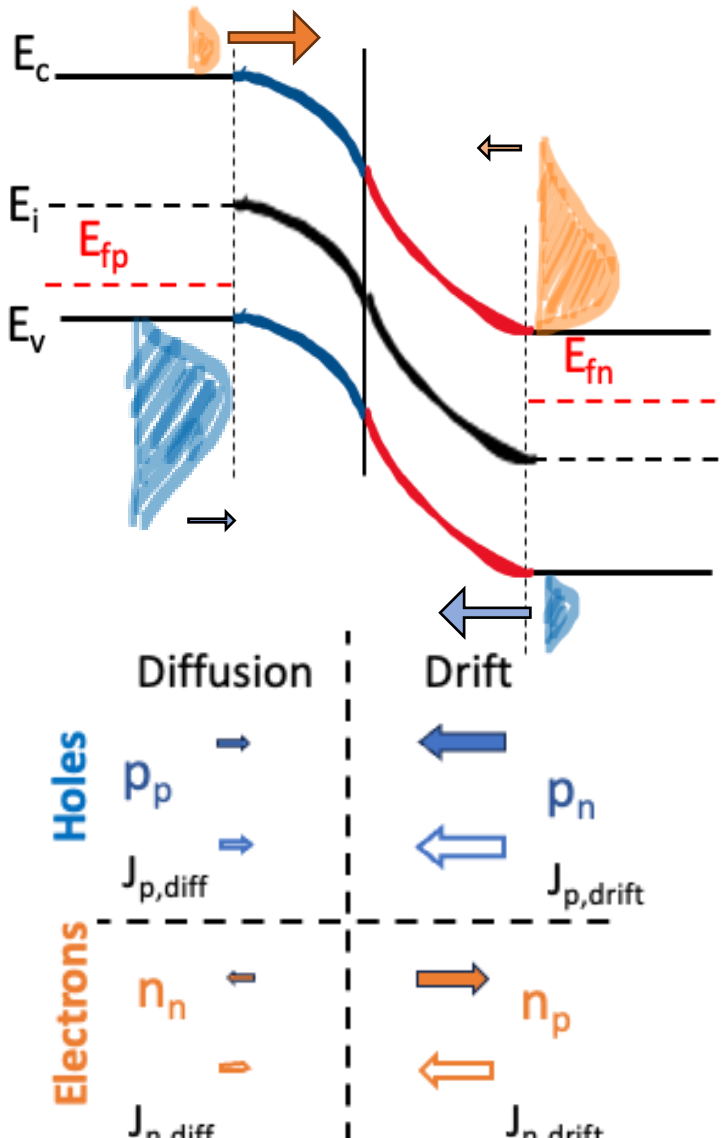
(a) Equilibrium: $V_{bias} = 0$



(b) FWD: $V_{bias} > 0$

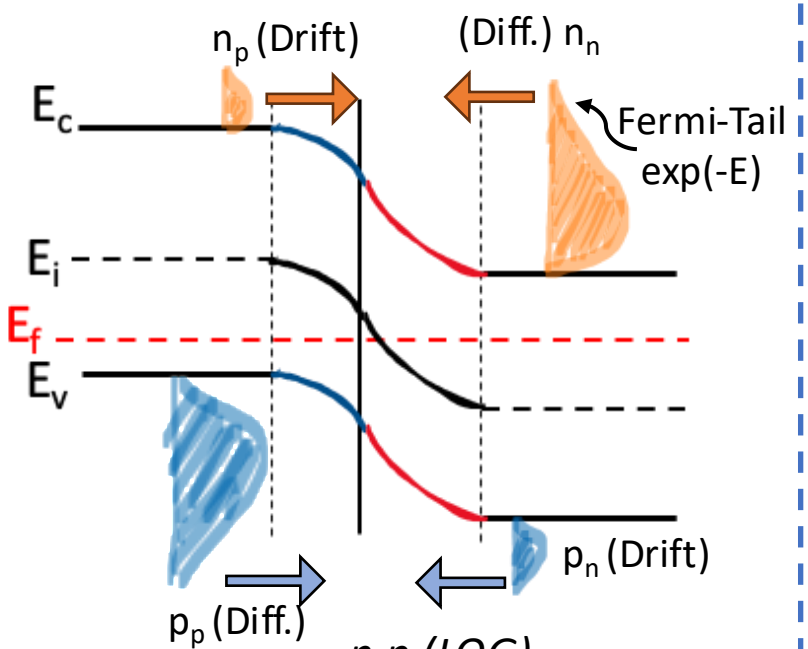


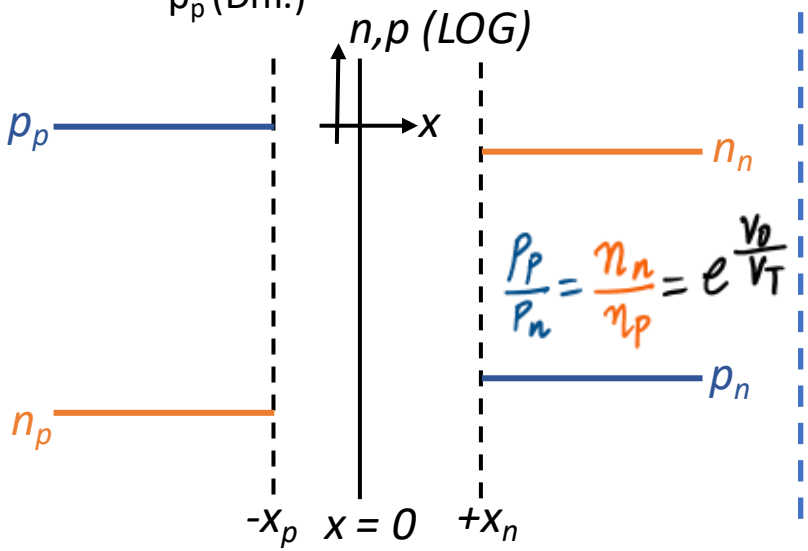
(c) REV $V_{bias} < 0$



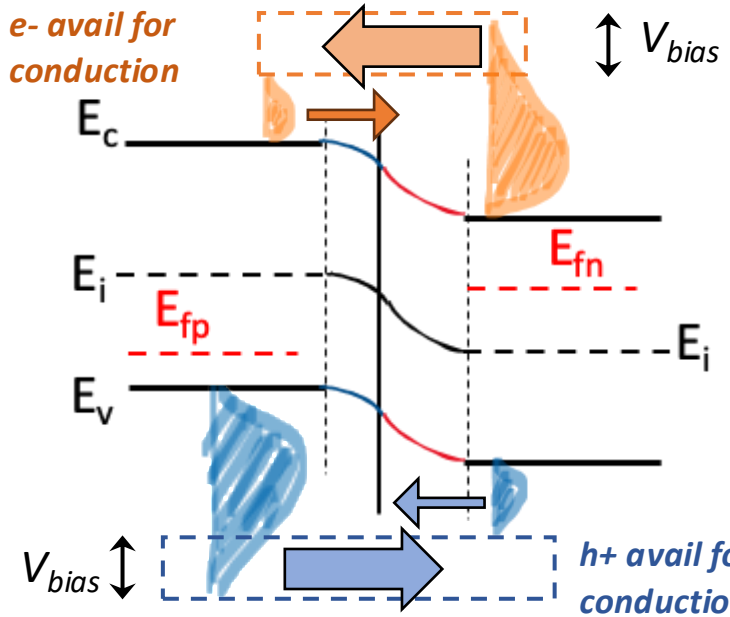
Non-Equilibrium Maj. and Min. Profiles

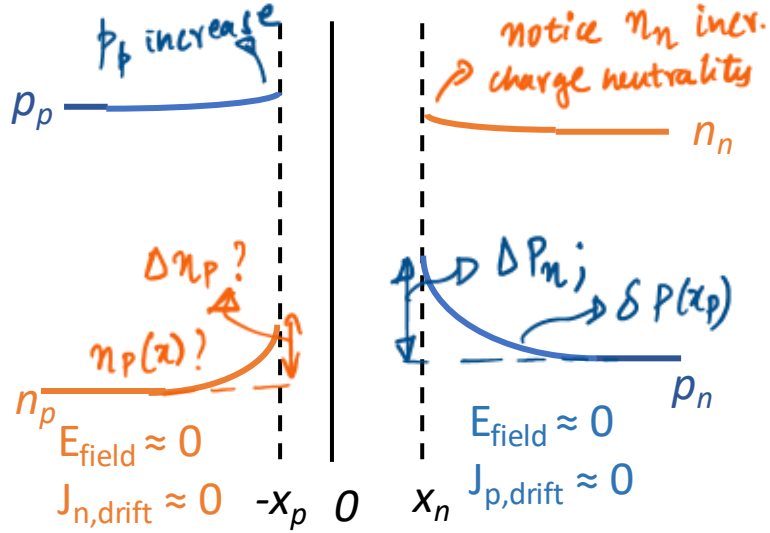
(a) Equilibrium: $V_{bias} = 0$



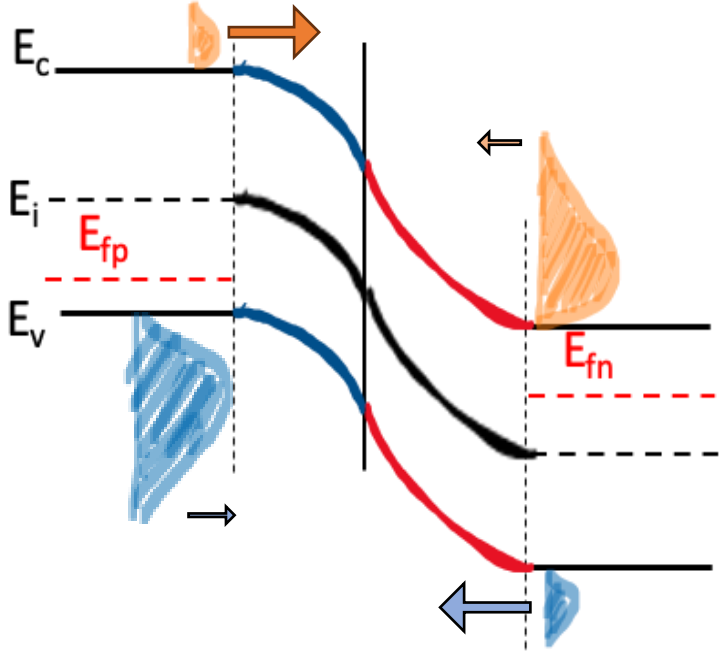


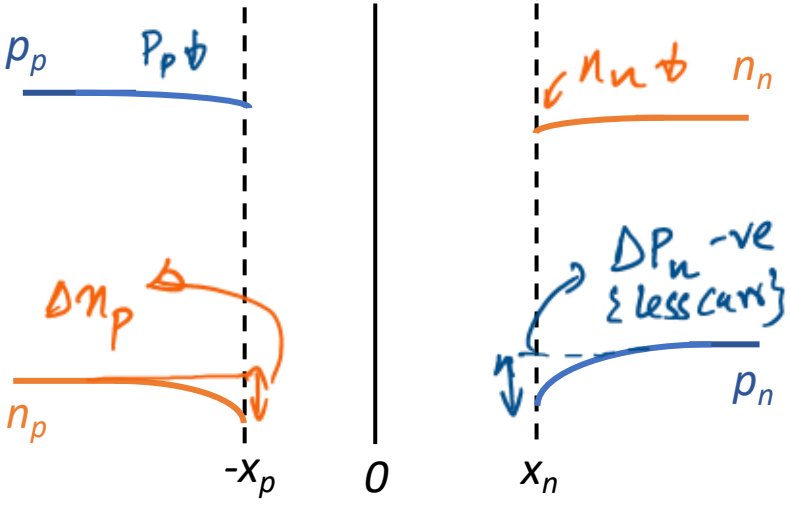
(b) FWD: $V_{bias} > 0$



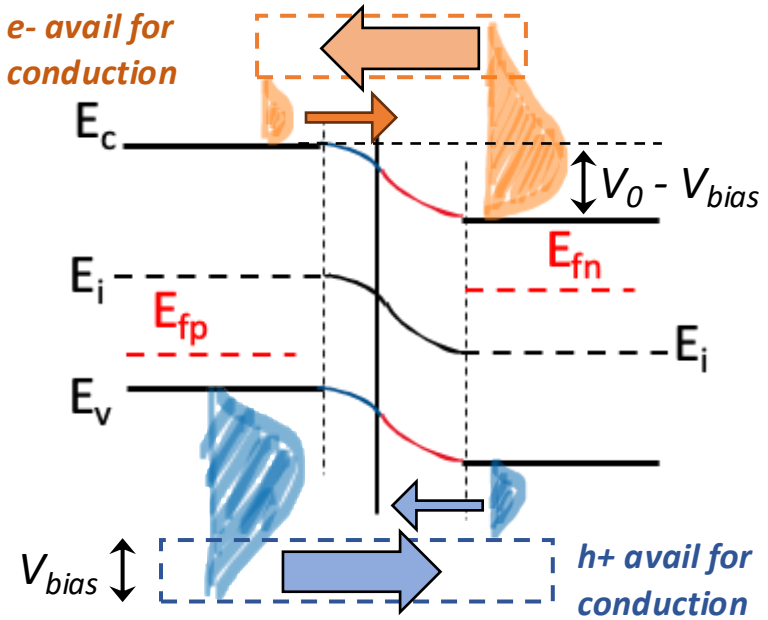


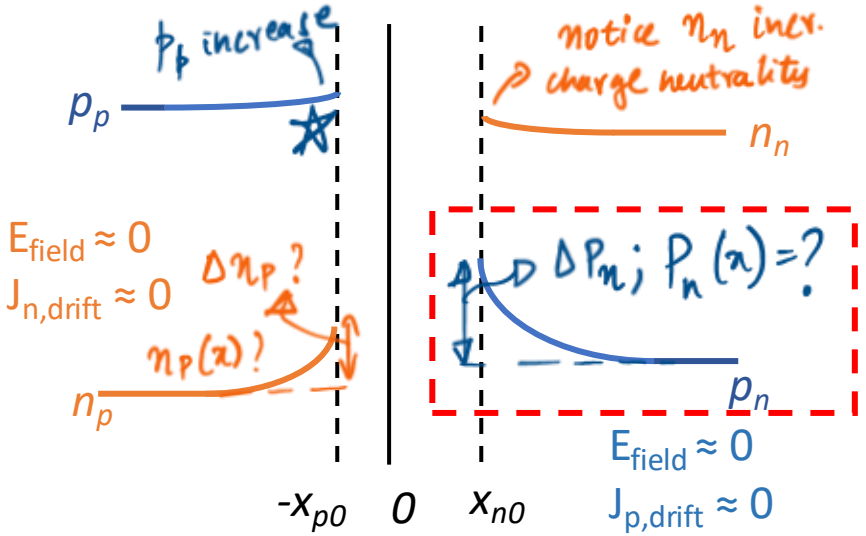
(c) REV $V_{bias} < 0$



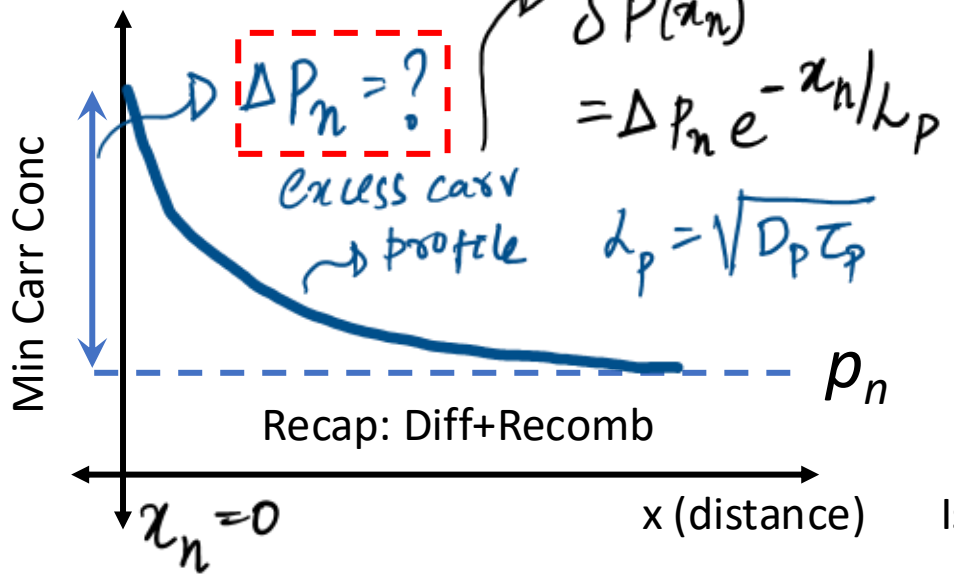


Minority Diffusion Current in Q.N. regions





Easiest to calculate $J_{p,diff}$ (x_{n+}) $J_{n,diff}$ (x_{p-})



① Equilibrium

$$\frac{p_p}{p_n} = e^{V_0/V_T} \quad \sim (1)$$

② FWD BIAS with INJ.

$$\frac{p(-x_{p0})}{p(+x_{n0})} = e^{\frac{(V_0 - V_{bias})}{V_T}} \quad \sim (2)$$

Is reasoning really, correct for Eqn. 2?

Puzzle 1: Derive this expression!

☆ Note :- Assuming low-level injection [LLI] :-

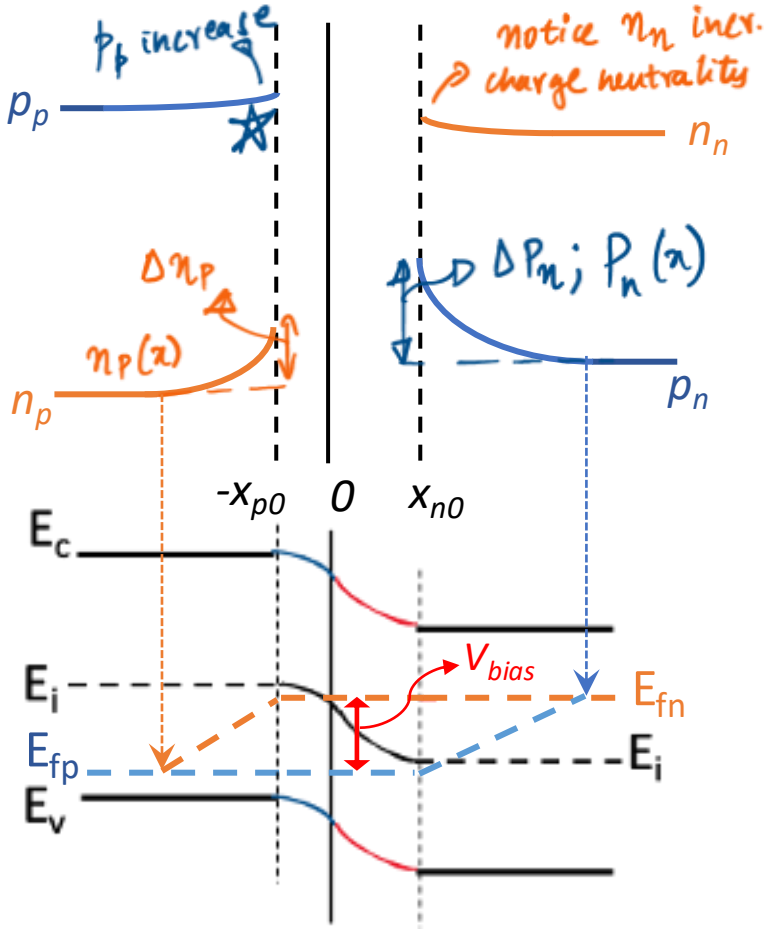
$\Delta n_p \ll p_p$; $\Delta P_n \ll n_n$, Therefore $p(-x_{p0}) \approx p_p$

Subs in ② & ① / ② ⇒ $p(x_{n0}) = p_n e^{V_{bias}/V_T}$

$$\Delta P_n = p(x_{n0}) - p_n = p_n \left(e^{\frac{V_{bias}}{V_T}} - 1\right) \quad (3.1)$$

Illy $\Delta n_p = n(x_{p0}) - n_p = n_p \left(e^{\frac{V_{bias}}{V_T}} - 1\right)$ (3.2)

Minority Diffusion Current in Q.N. regions



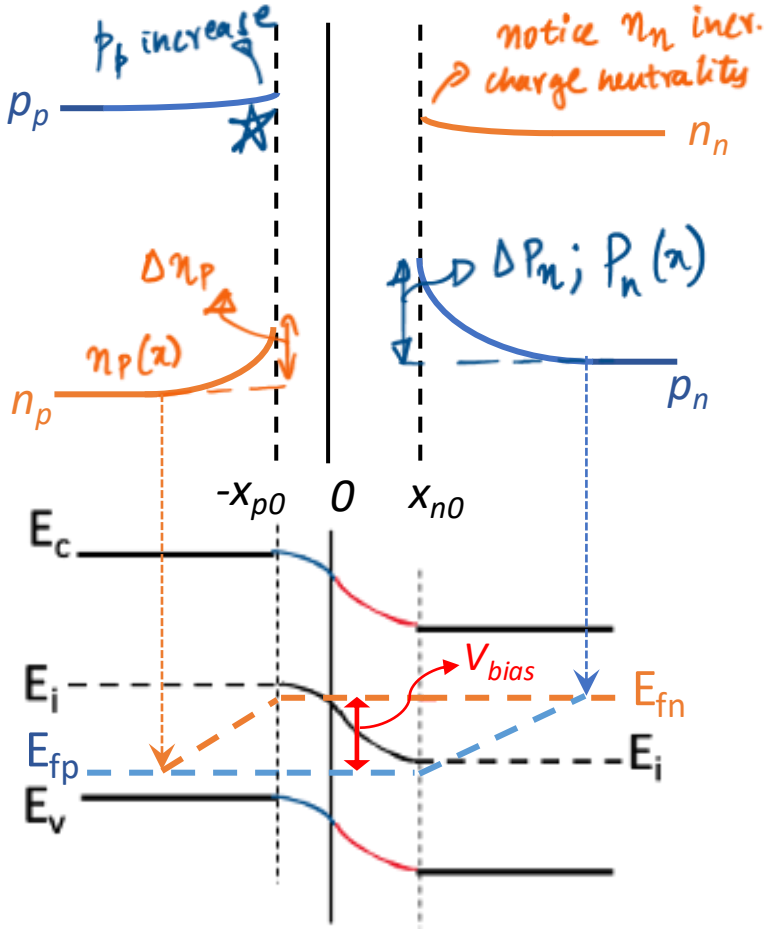
Band Diagram with Complete E_{fn}, E_{fp}

Assumption: No Recomb in Depletion

1. Maj Carr & $\nabla E_{Fp,n} \approx 0$
2. Min Carr exp decay => $E_{Fp,n}$ linear n,p
3. $E_{Fp,n}$ merge where δn_p ,$\delta p_n \approx 0$

Hint: Puzzle 1 Derivation in Band Diagram!

Minority Diffusion Current in Q.N. regions

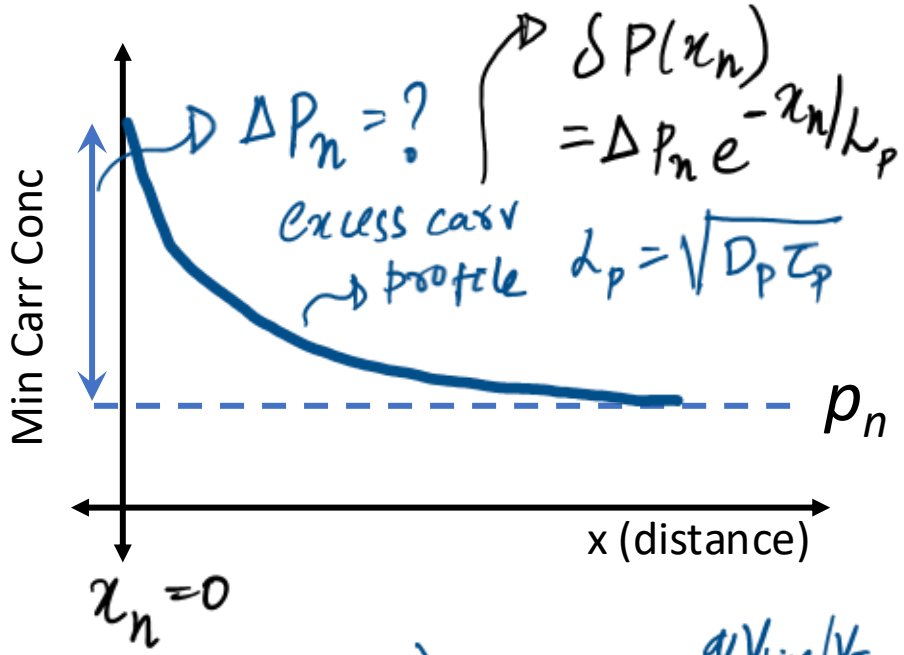


Band Diagram with Complete E_{fn}, E_{fp}

Assumption: No Recomb in Depletion

1. Maj Carr & $\nabla E_{Fp,n} \approx 0$
2. Min Carr exp decay => $E_{Fp,n}$ linear n,p
3. $E_{Fp,n}$ merge where $\delta n_p, \delta p_n \approx 0$

Hint: Puzzle 1 Derivation in Band Diagram!



$$\Delta P_n = p_n\left(e^{\frac{V_{bias}}{V_T}} - 1\right) \quad (3.1)$$

$$\Delta n_p = n_p\left(e^{\frac{V_{bias}}{V_T}} - 1\right) \quad (3.2)$$

Since, Excess carr $\delta P(x_n) = \Delta P_n e^{-x_n/L_p}$

$$\delta p(x_n) = p_n\left(e^{qV_{bias}/V_T} - 1\right) e^{-x_n/L_p} \quad (4.1)$$

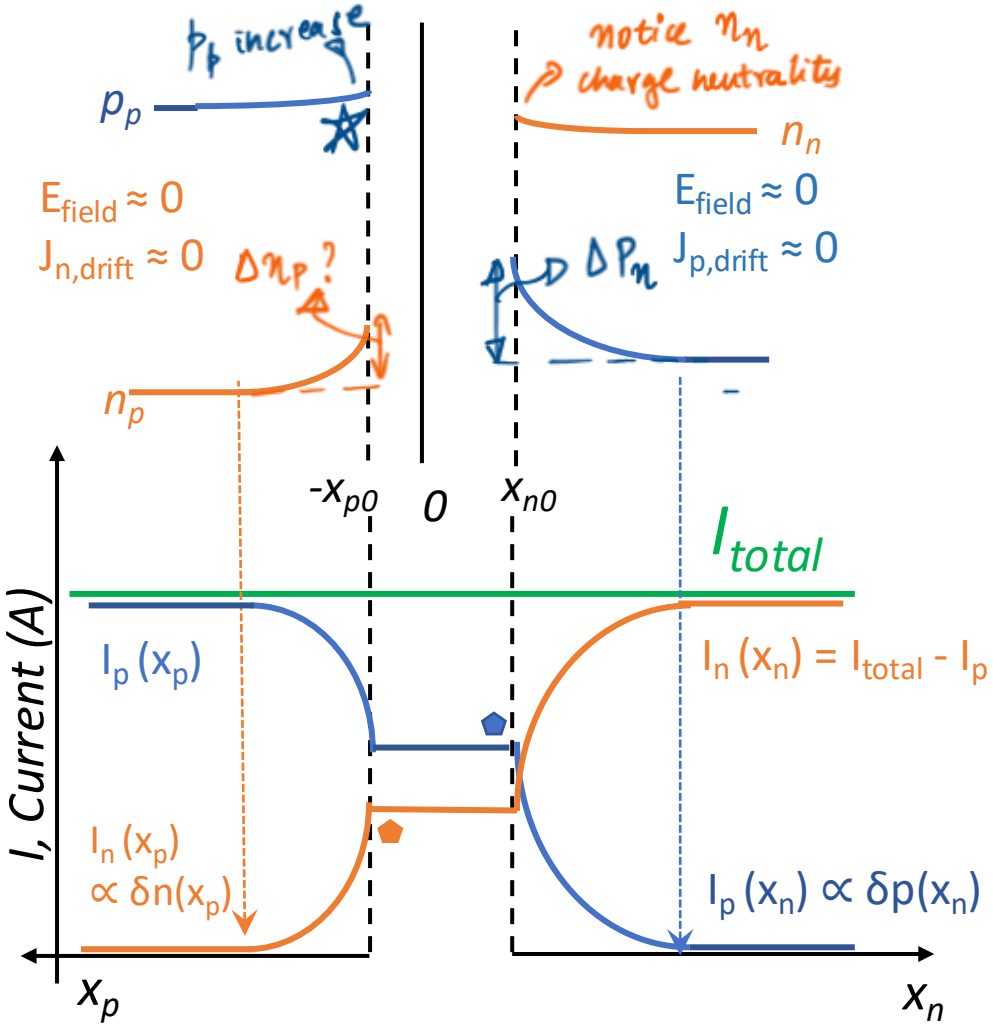
$$\text{III}^{ly}\ \delta n(x_p) = n_p\left(e^{qV_{bias}/V_T} - 1\right) e^{-x_p/L_n} \quad (4.2)$$

Since $I_{diff,p} = -qD_p A \frac{d\delta P(x_n)}{dx_n} = qA\frac{D_p}{L_p}\delta p(x_n)$ (5.1)

$$\text{III}^{ly}\ I_{diff,n} = -qA\frac{D_n}{L_n}\delta n(x_p) \quad (5.2)$$

{Notice sign of $I_{diff,p}$ & $I_{diff,n}$}

I_{total} and $I_{p,n}(x)$



We know $I_p(x_n)$ and $I_n(-x_p)$, But I_{total}?

Puzzle 2: Prove $\nabla I_{total}(x) = 0$ ***Hint: Steady State!***

In Depletion: $\nabla I_{p,n}(x) \approx 0$ -> NO net Recomb/Gen

$$I_{total} = I_p(x_n=0) - I_n(x_p=0)$$

$$= qA\frac{D_p}{L_p}\Delta P_n + qA\frac{D_n}{L_n}\Delta n_p$$

$$= qA\left[\frac{D_p}{L_p}P_n + \frac{D_n}{L_n}n_p\right]\left[e^{V_{bias}/V_T} - 1\right]$$

↳ Shockley Diode Current Eqn.

note: $\because P_n > n_p \Rightarrow p_p > n_n$

↳ I dominated by heavily doped side

$$I(V_{bias}) = I_0\left[e^{V_{bias}/V_T} - 1\right]$$

1) FWD $+V_{bias} >$ few $V_T \Rightarrow I_{FWD} = I_0 e^{V_{bias}/V_T}$

2) REV $-V_{bias} >$ few $V_T \Rightarrow I_{REV} = -I_0$

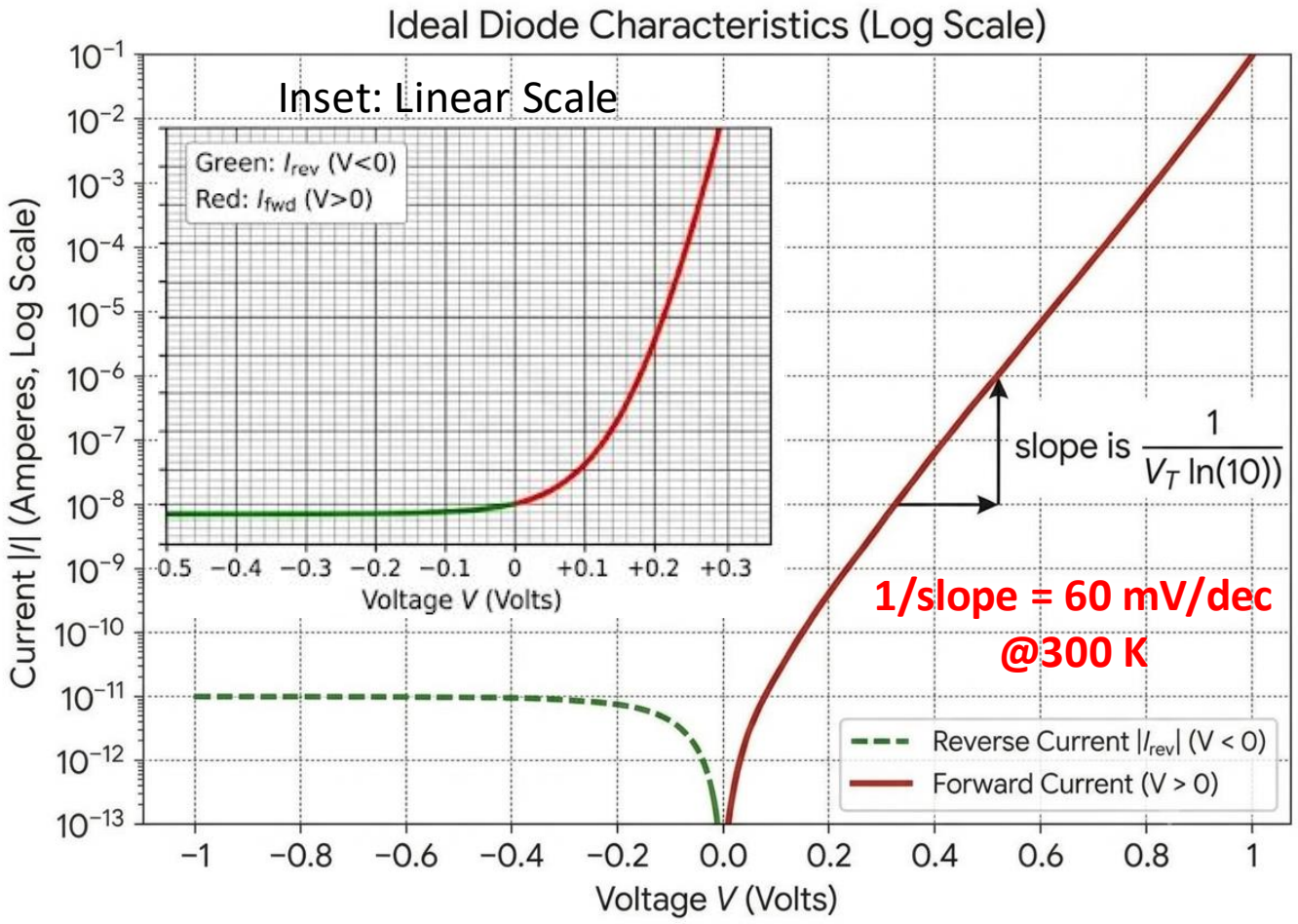
$-I_0 \triangleright$ rev saturation current $\Rightarrow$ drift of minority carr.

Puzzle 3: What is the nature of Maj. Current?

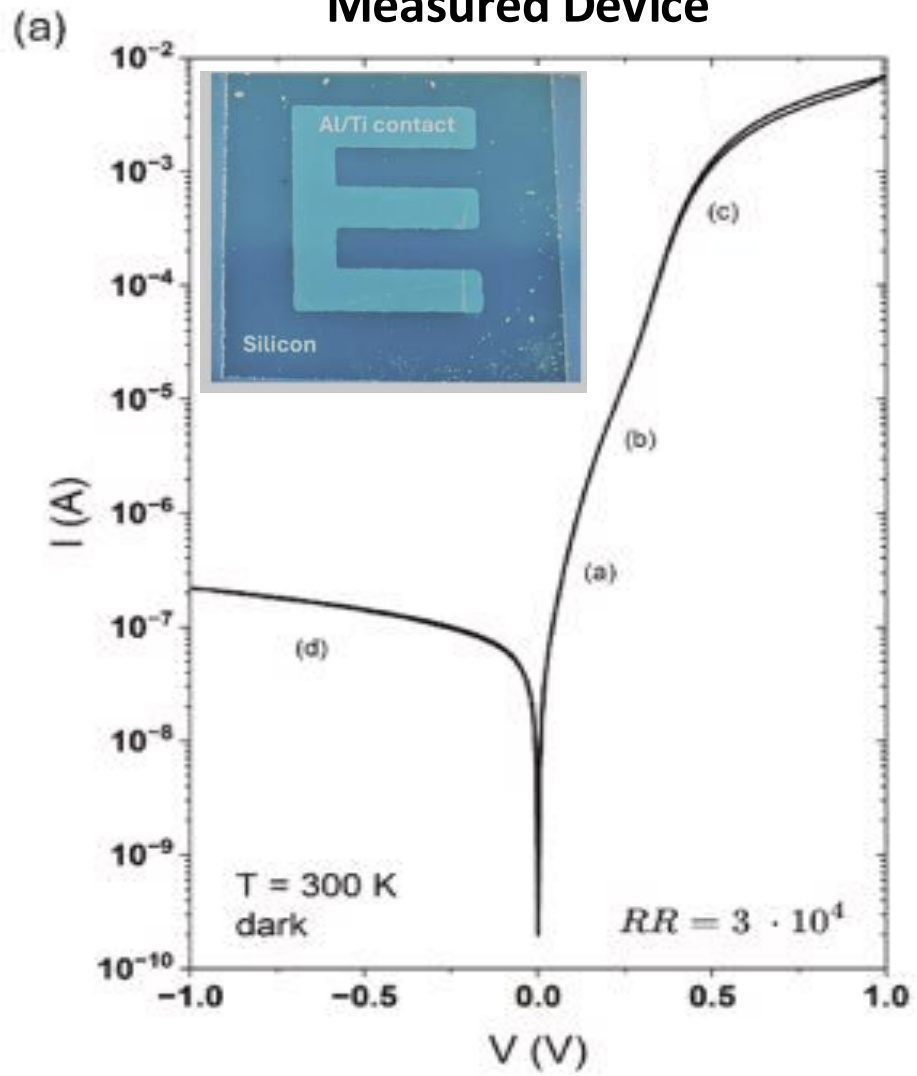
Puzzle 4: Maj. Carr $J_n = q\mu_n \nabla E_{fn}$ J_n large but $\nabla E_{fn} \approx 0$

Ideal Diode Characteristics – in Linear and log scale

Ideal Shockley Diode: $I(V_{bias}) = I_0\left[e^{V_{bias}/V_T} - 1\right]$

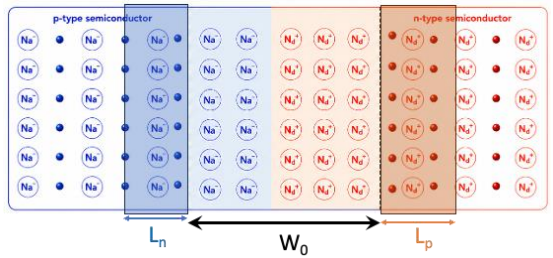


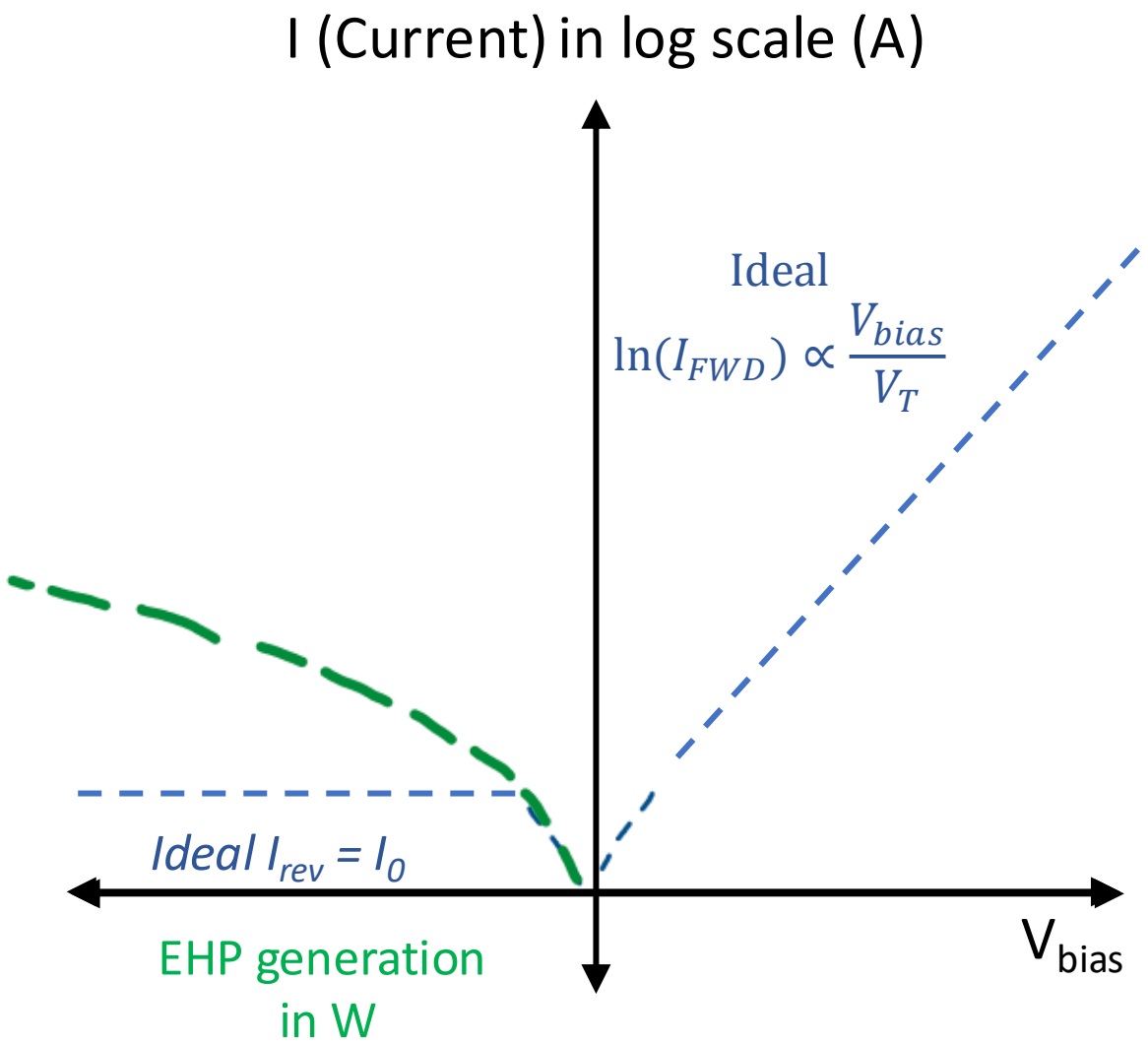
Nonidealities in Measured Device



Source: 10.1016/j.physb.2026.418479

Non-Idealities in I-V Characteristics





1. REV SAT Current (I_0)

- Assumption: n_p,p_n generated within $L_{n,p}$ of Q.N. regions (Depletion edge)
- If moderate doping, W -> large; W grows in V_{rev}
- Significant G_{th} in W -> swept away before recombining! Deviation from I_0

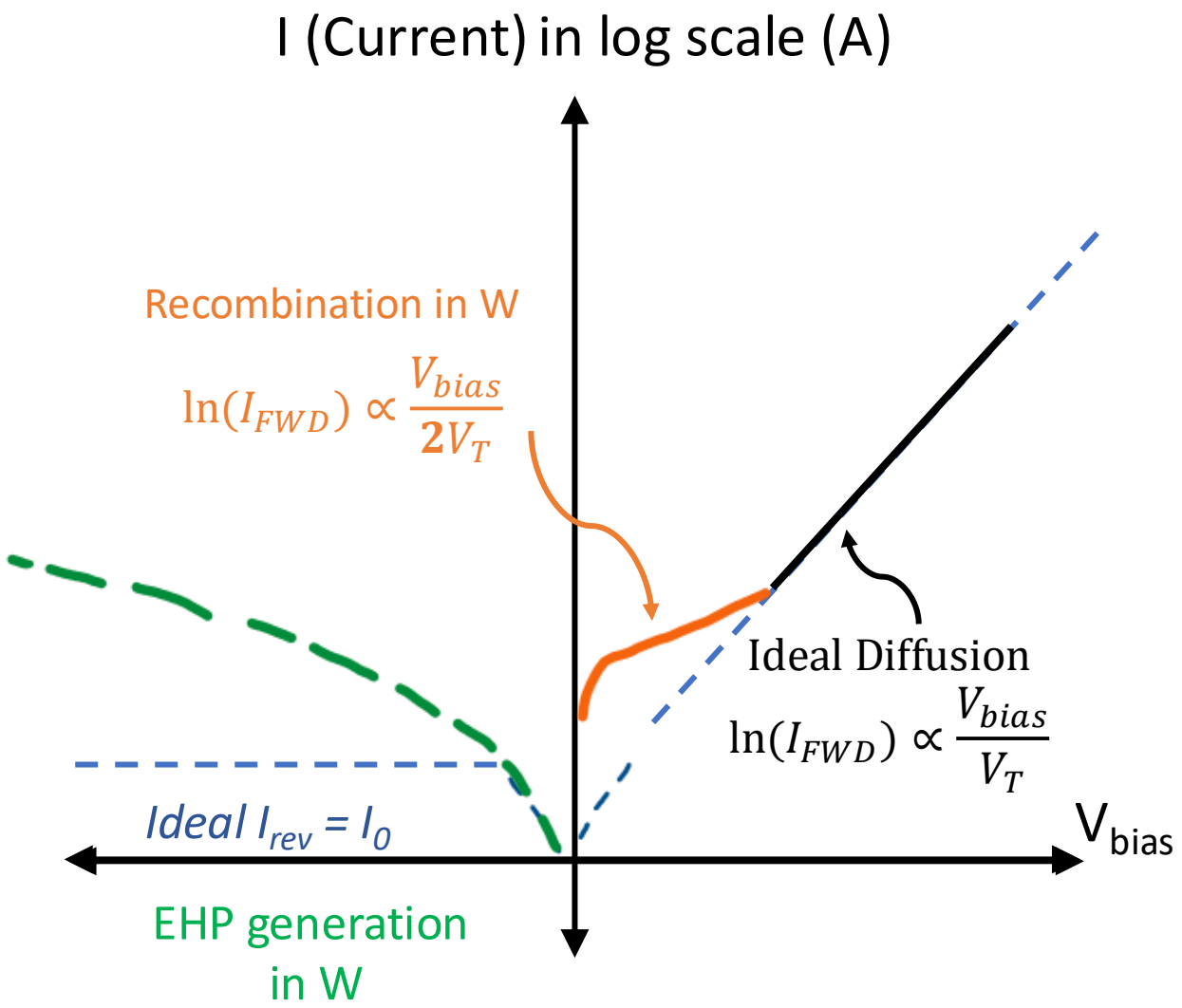
SRH Generation: $G_{th} \approx \frac{n_i}{2\tau_0}$ $J_{gen} = q\int_0^W G_{th}\, dx = \frac{qn_iW}{2\tau_0}$

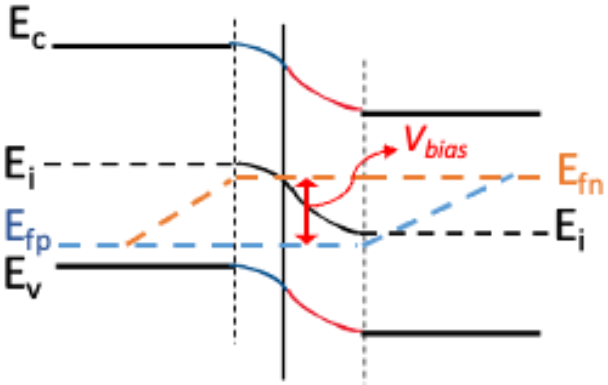
$$W(V_{REV}) \propto \sqrt{V_{bi} + |V_{REV}|}$$

$$J_{reverse} = J_0 + \frac{qn_iW(V_{REV})}{2\tau_0}$$

- *Does not affect FWD bias – Why?*

Non-Idealities in I-V Characteristics





2. FWD: LOW V_{bias}

- Assumption: No recombination in space charge region. $\tau_{rec} >> \tau_{transit}$
- However, large *n,p* meet inside "W"

$$n \cdot p = n_i^2 e^{(E_{fn}-E_{fp})/kT} = n_i^2 e^{qV_{bias}/kT}$$

- Max Recombination rate,

$$@\ n \approx p \approx n_i e^{qV/2kT}:$$

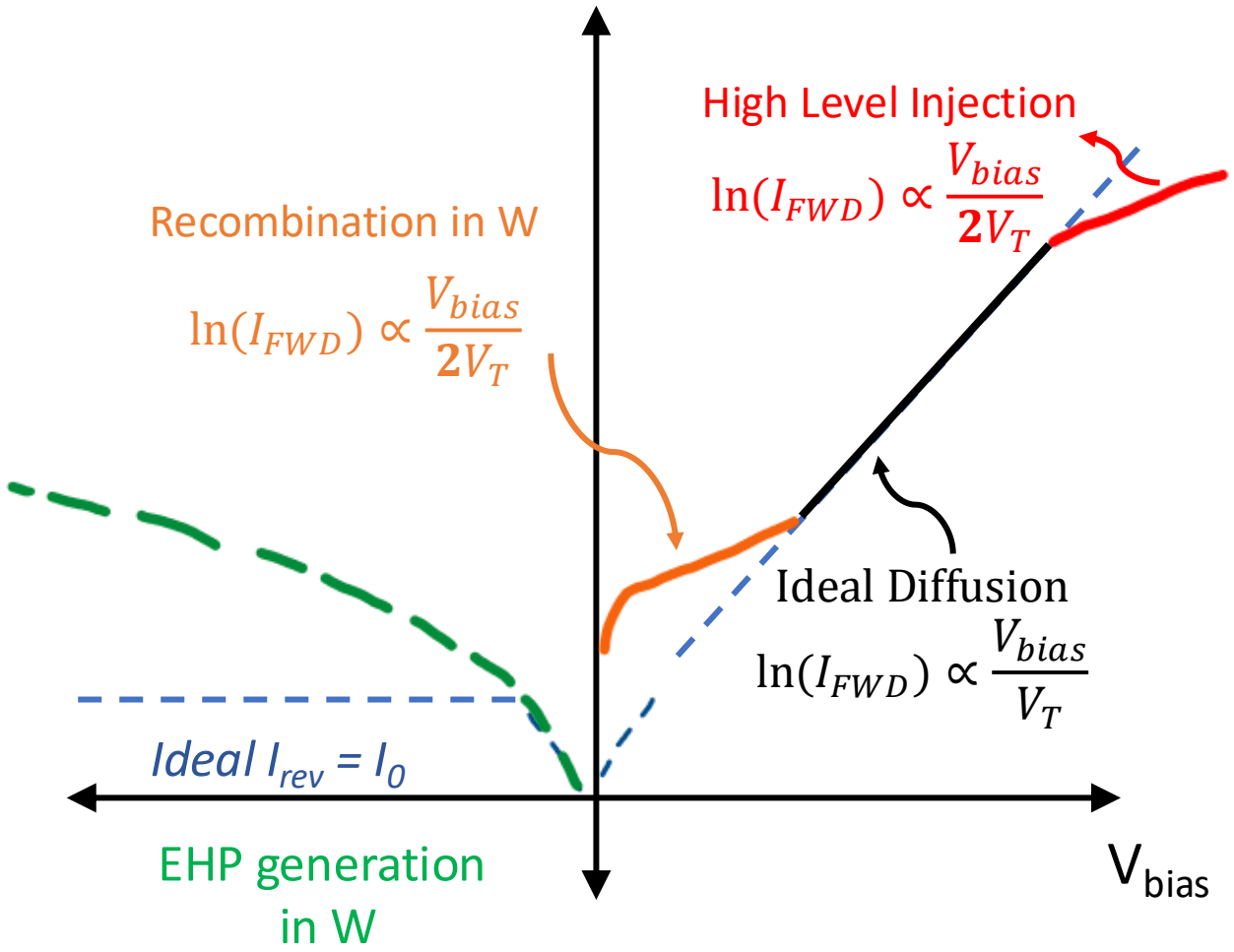
$$R_{max} \approx \frac{n \approx p}{2\tau_0} \approx \frac{n_i}{2\tau_0} e^{\frac{qV}{2kT}}$$

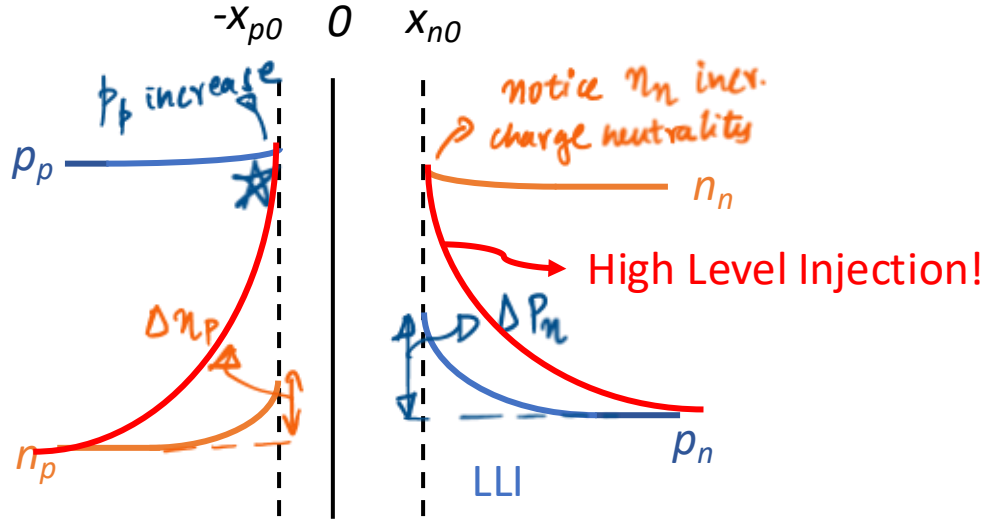
$$J_{rec} = q \int_0^W U\ dx \approx \frac{qWn_i}{2\tau_0} e^{\frac{qV}{2kT}} = J_{0,rec}\, e^{\frac{V}{2V_t}}$$

- V_{bias} > 0.1 - 0.2 V -> J_{diff} >>J_{rec}

$$J_{diff} = J_{0,diff}\left(e^{\frac{V}{V_t}} - 1\right)$$

Non-Idealities in I-V Characteristics





3. FWD: HIGH V_{bias}

Ideal, Low V_{bias}: Low level Inj:

$\Delta n_p << p_p$ and $\Delta p_n << n_n$

$J_{diff} \propto \Delta p_n, \Delta n_p \propto e^{qV_{bias}/kT}$

High V_{bias}: High level Injection

$\Delta n_p \approx p_p$ & $\Delta p_n \approx n_n$,

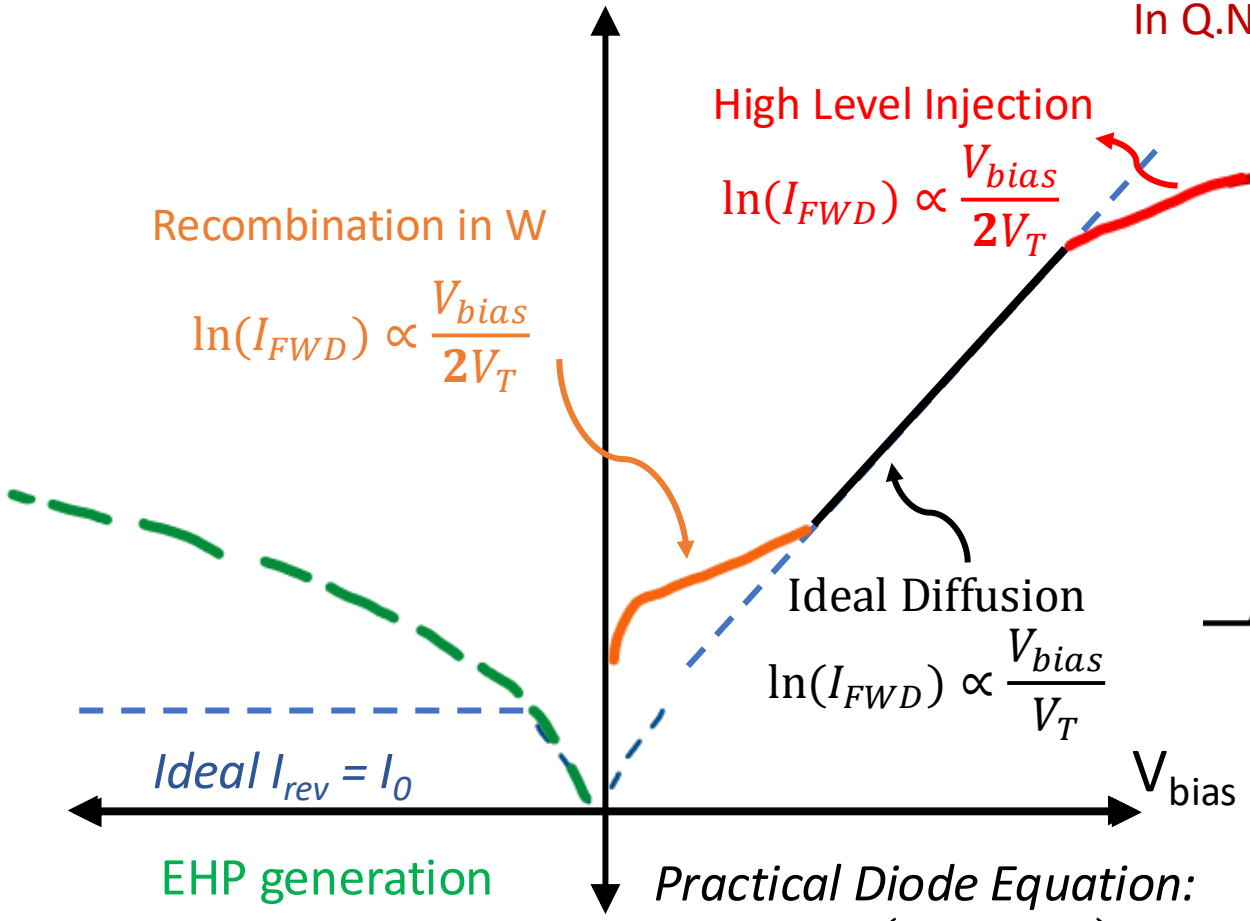
Consider @ x_{n0} : $np \approx n_i^2 e^{qV_{bias}/kT}$

$n_n \Delta p_n \approx (\Delta p_n)^2 \approx n_i^2 e^{qV_{bias}/kT}$

Δp_n & $\Delta n_p \approx n_i e^{qV_{bias}/2kT}$

$J_{diff} \propto e^{qV_{bias}/2kT}$ (@ High level Inj.)

Non-Idealities in I-V Characteristics



Practical Diode Equation:

$$I = I_0\left(e^{\frac{qV}{\eta kT}} - 1\right)$$

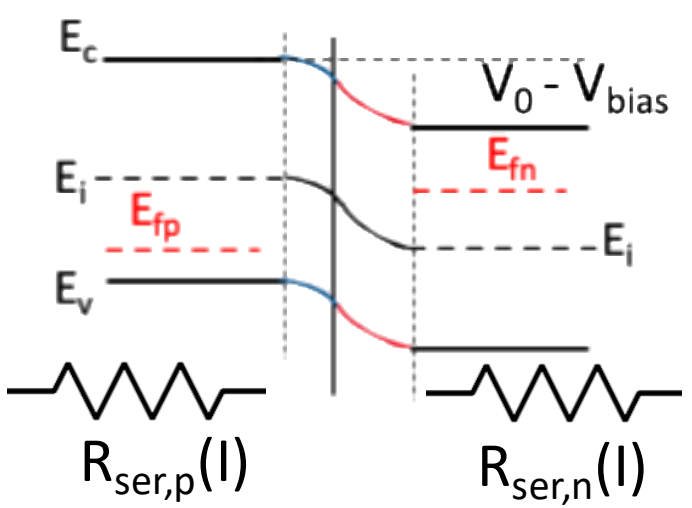
$\eta \approx 1 - 2$ (Non-Ideality Factor)

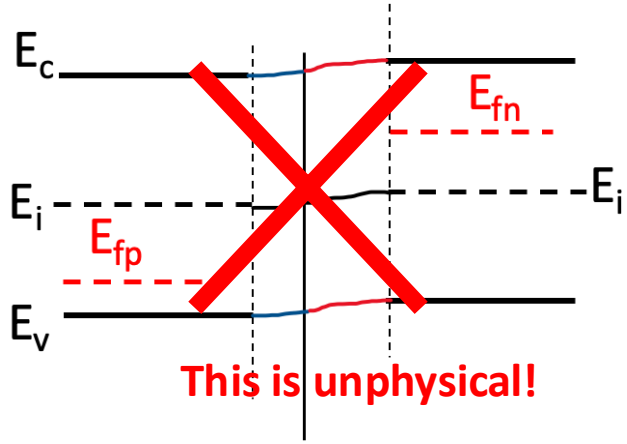
4. FWD: Very HIGH V_{bias}

Q: Let's look at Band Diagram of FW *pn* junction

What happens when $V_{bias} \geq V_0$

Flip in Bands & Direction of E_{field}??





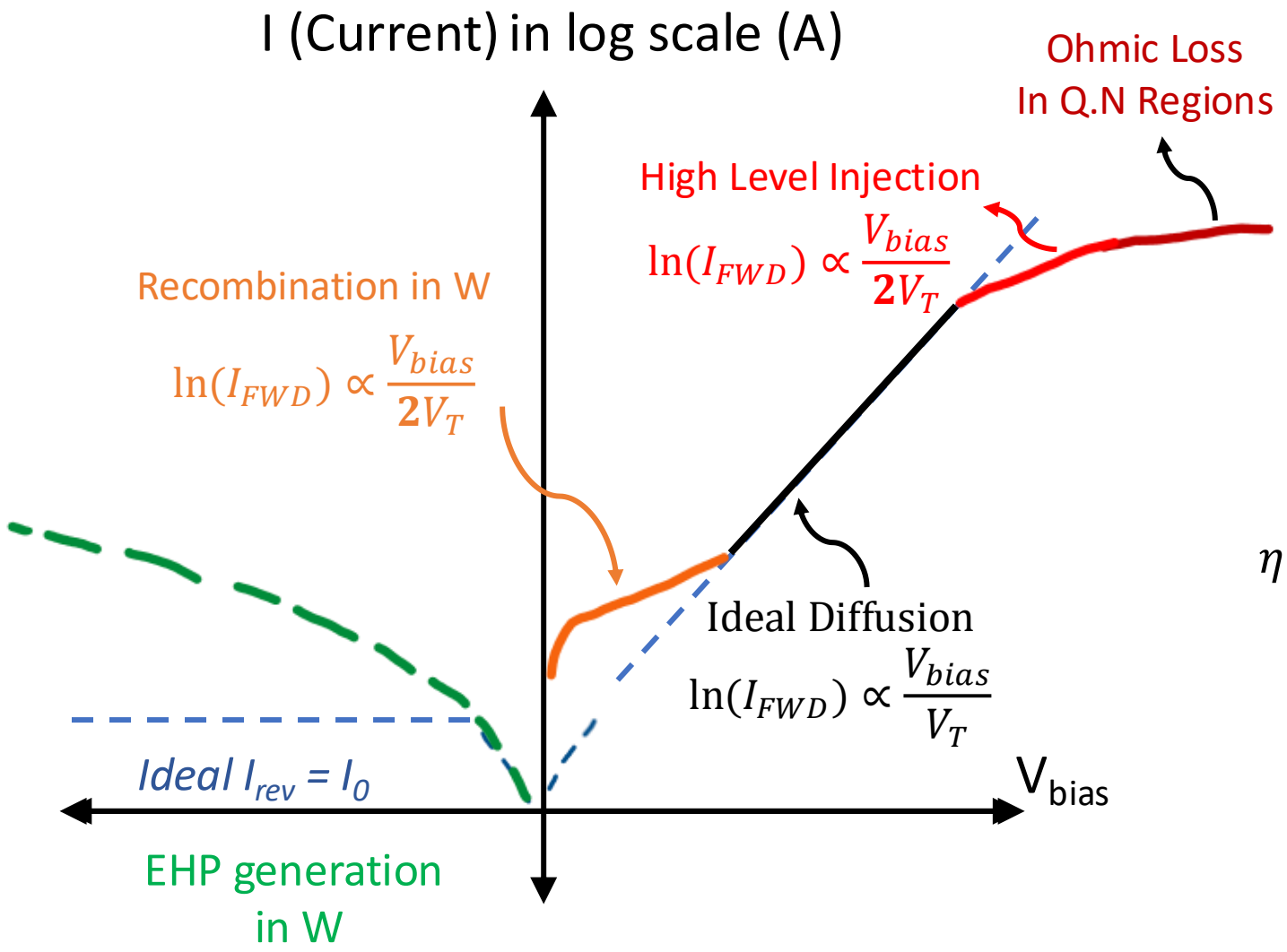
Assumption all V_{bias} appears on Depletion Region

At high V_{bias}, I – not true– Ohmic loss, heating

Significant drop across Q.N p and n regions

$$V_{bias} = V_{junc} + I[R_{ser,p}(I) + R_{ser,n}(I)]$$

Non-Idealities in I-V Characteristics - Summary

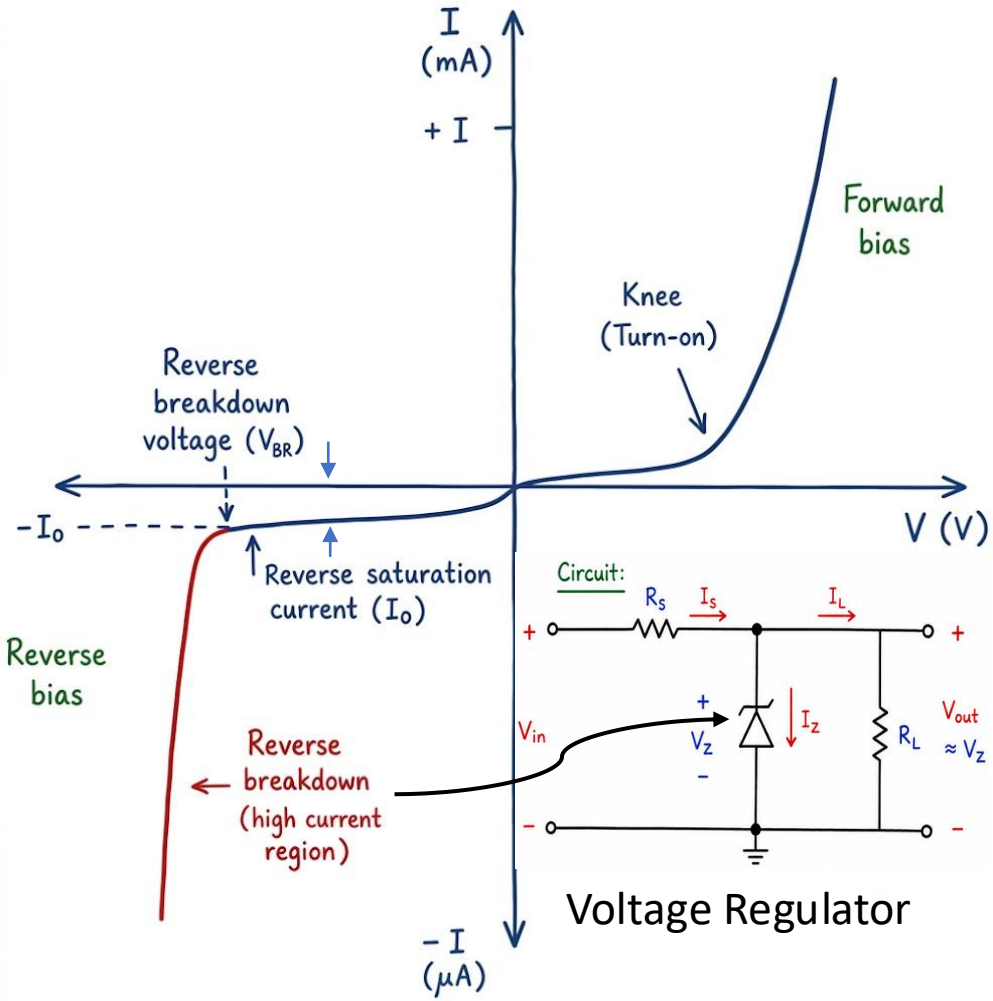


Practical Diode Equation:

$$I = I_0\left(e^{\frac{qV}{\eta kT}} - 1\right)$$

$\eta \approx 1 - 2$ (Non-Ideality Factor)

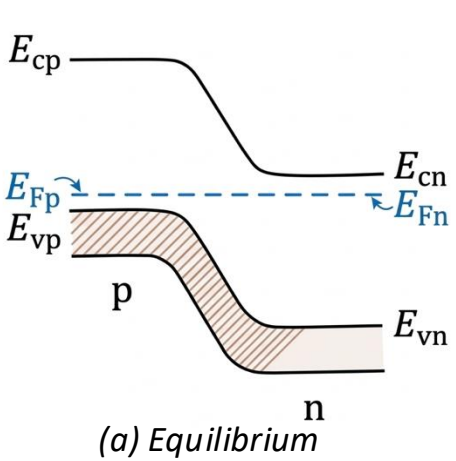
Reverse Bias Breakdown (2 Mechanisms)



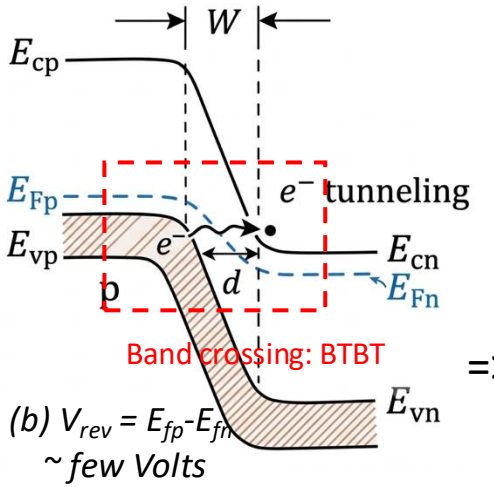
Voltage Regulator

"Breakdown" – sudden increase $J_{rev}(V_{br}+)$
Nothing destructive- But! limit current (R_{ser})
Application: Voltage Regulator

1. Zener Breakdown



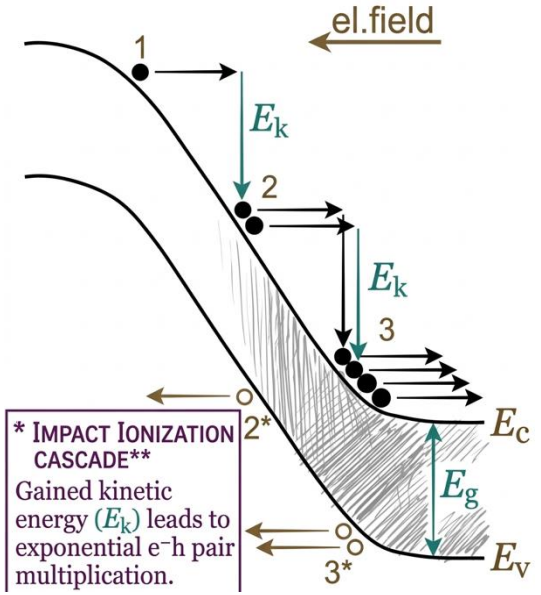
(a) Equilibrium



(b) $V_{rev} = E_{fp}-E_{fn}$
~ few Volts

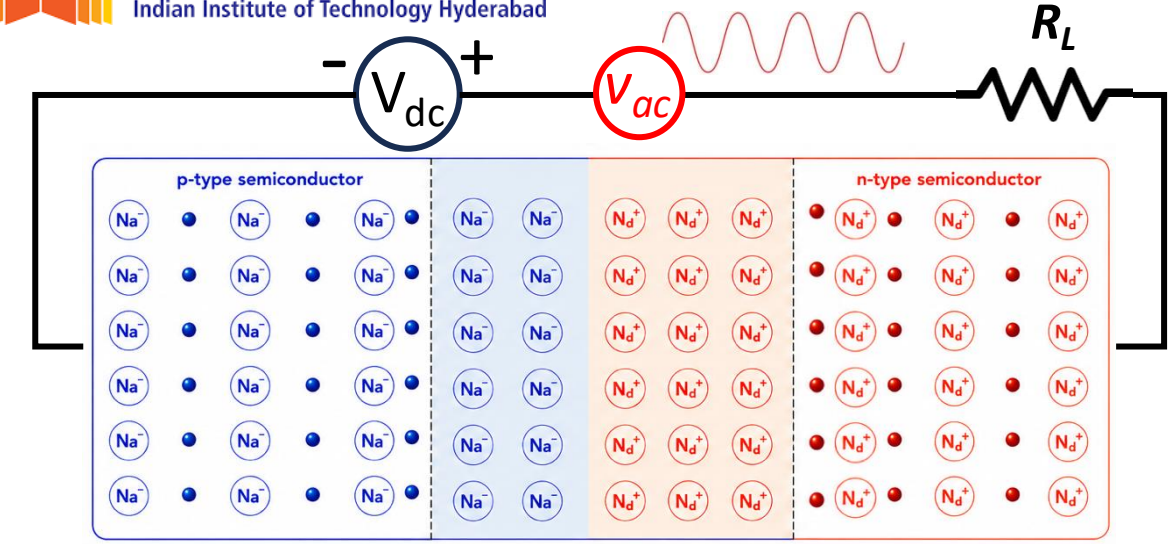
Heavily doped p+/n+ junctions:
(a) Thin *W < 10 nm*
- large p_p, n_n
(b) "Band Crossing" at few Volts of V_{rev}
=> Band-to Band Tunneling!
Large *I at* V_{br}

2. Avalanche Breakdown

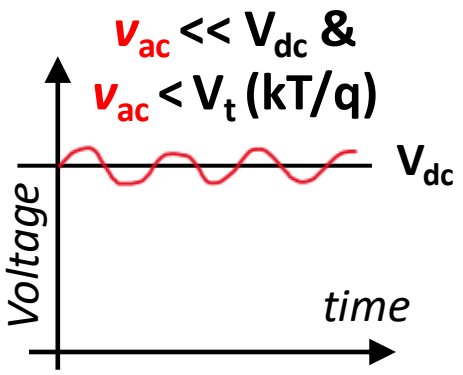


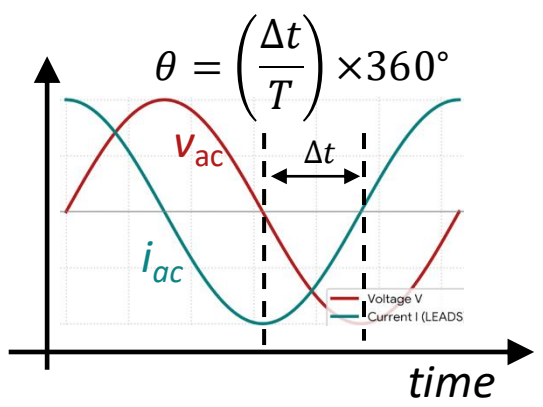
- Light to moderate p/n:
- Large V_{br} (10-1000 V)
- Large E_{br} -> Carr acceleration
- High K.E: Impact ionization
- Chain reaction: Avalanche effect
- Carr MULT, large current

C-V Characteristics – Depletion/Junction Capacitance

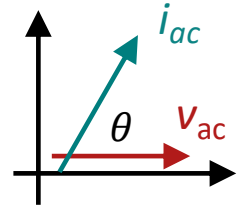


Small Signal Analysis





Phasor

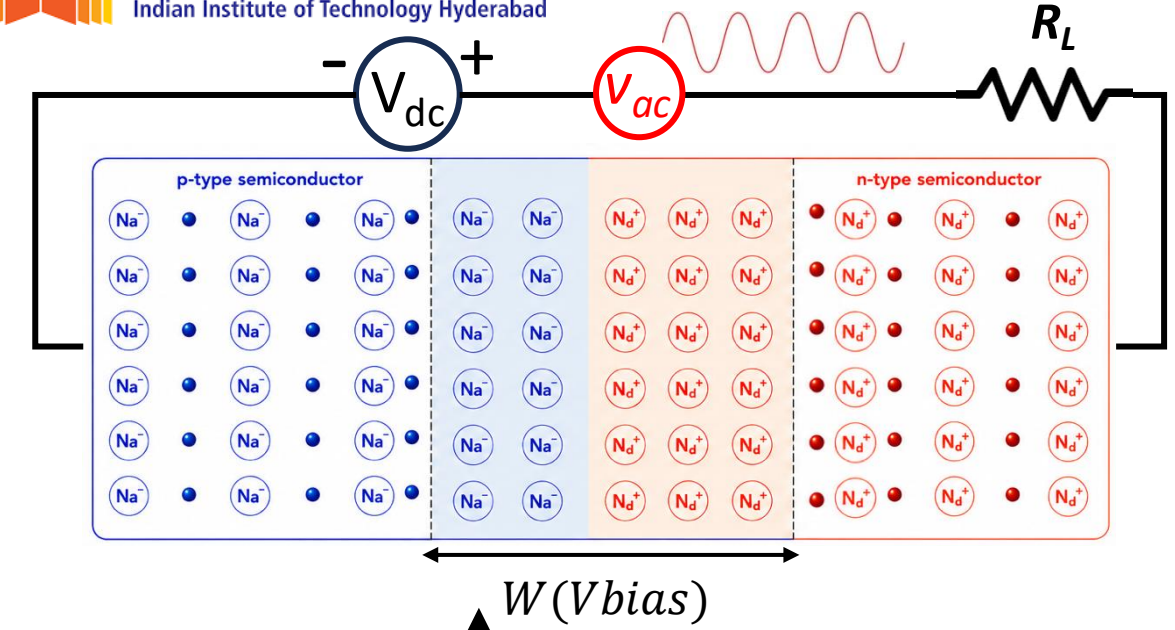


$$\tan\theta = \frac{X_C}{R_L} = \frac{1}{j\omega C R_L}$$

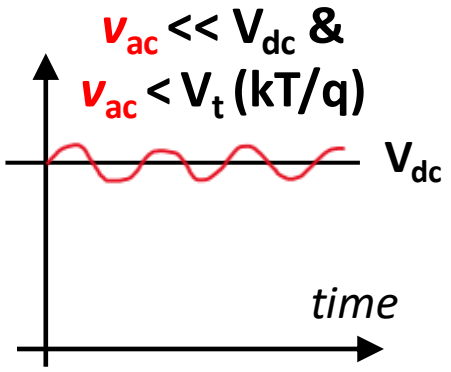
Why should a diode exhibit Capacitance?
How to measure it?

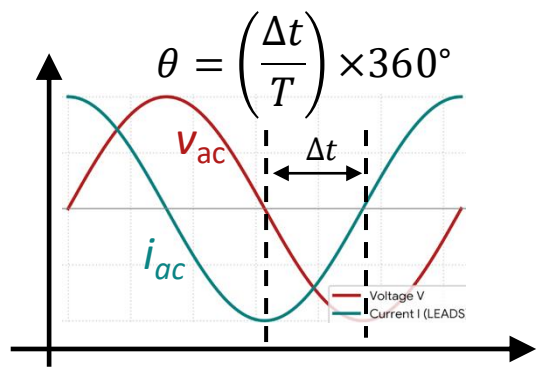
भारतीय सांकेतिक विज्ञान संस्थ हैदराबाद
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

C-V Characteristics – Depletion/Junction Capacitance

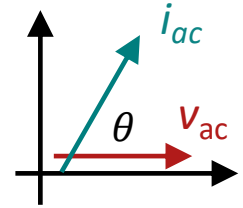


Small Signal Analysis



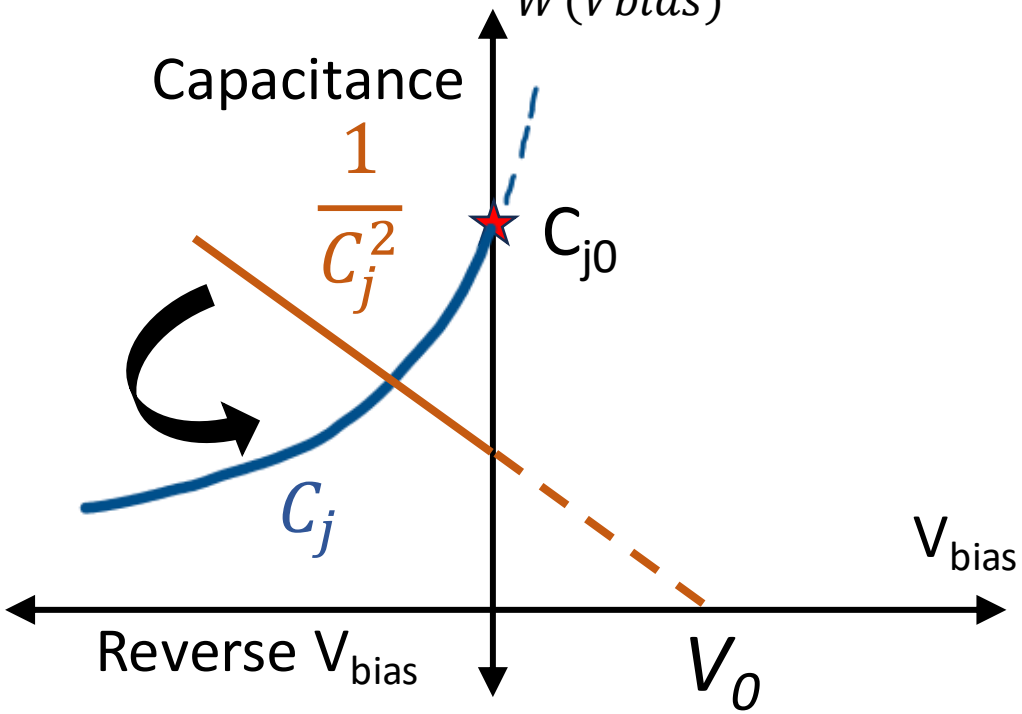


Phasor



$$\tan\theta = \frac{X_C}{R_L} = \frac{1}{j\omega C R_L}$$

$$C_j = \left|\frac{dQ_{dep}}{dv_{ac}}\right| \rightarrow dQ_{dep} = qAN_A dx_p = qAN_D dx_n$$



Parallel Plate Approx:

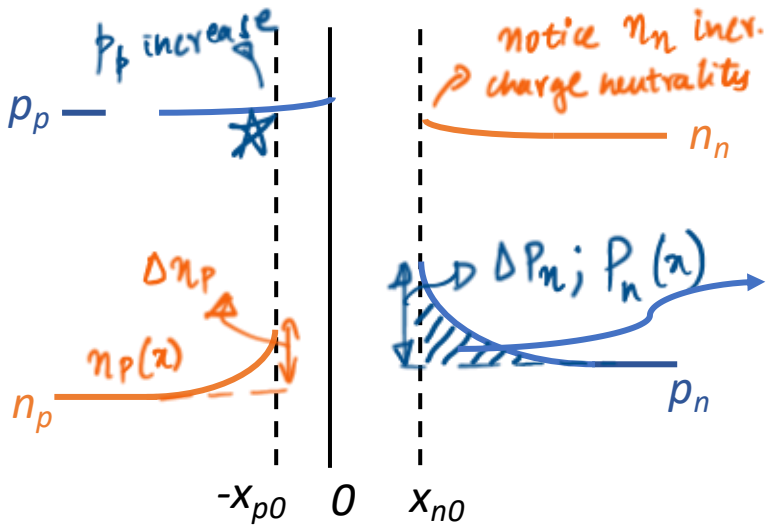
$$C_j = \frac{\varepsilon_S A}{W(Vbias)} = A\sqrt{\frac{q\varepsilon_S}{2(V_0+|V_{bias}|)}\left(\frac{N_A N_D}{N_A+N_D}\right)}\ ;\ V_{bias} < 0$$

$$C_{j0} = \frac{\varepsilon_S A}{W_0}\ ;\ V_{bias} = 0$$

$$\frac{1}{C_j^2} = \frac{2(V_0 + V_{bias})}{q\varepsilon A^2}\left(\frac{1}{N_A} + \frac{1}{N_D}\right)$$

Intercept: $1/C_j^2$ vs V_{bias} gives V_0 ; **Slope** $\propto \frac{1}{N_{smaller}}$)

C-V Characteristics – Diffusion Capacitance

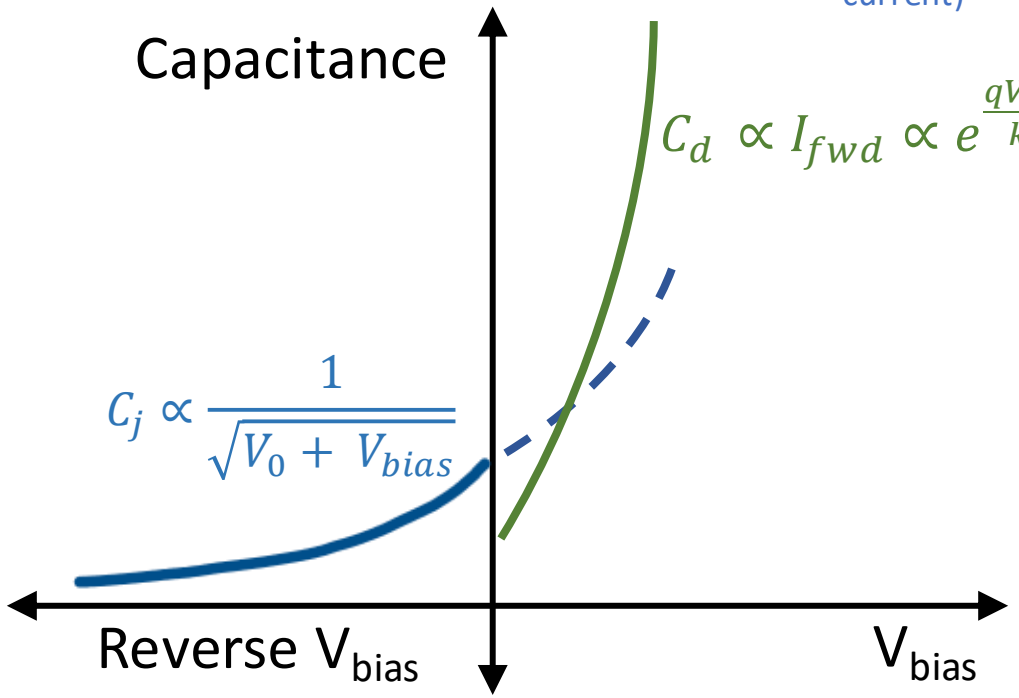


Stored diffusion minority charge

Area under the curve

$Q_{stored} \approx I_{fwd} \cdot \tau_p$

Puzzle 5: Prove above expression (Another derivation for diode current)



Diffusion Capacitance (C_d) — Forward Bias

Diffused (injected) minority carriers stored in (ΔQ_{stored}) in Q.N. regions. v_{ac} modulates this stored minority charge.

$$I_{fwd}(t) = I_0\left(e^{\frac{V_{DC}+v_{ac}(t)}{V_t}} - 1\right) \approx I_0 e^{\frac{V_{DC}}{V_t}} \cdot e^{\frac{v_{ac}(t)}{V_t}} = I_{DC} \cdot e^{\frac{v_{ac}(t)}{V_t}}$$

Taylor series expansion $e^{\frac{v_{ac}(t)}{V_T}}$

Ignore O(>1) if $v_{ac} << V_t$

$$I_{fwd}(t) = I_{DC}\left[1 + \frac{v_{ac}(t)}{V_t} + \frac{1}{2}\left(\frac{v_{ac}(t)}{V_t}\right)^2 + \frac{1}{6}\left(\frac{v_{ac}(t)}{V_t}\right)^3 + \cdots\right]$$

Linear term

C_d change in stored minority carrier charge with applied voltage:

$$C_d = \frac{dQ_{min,stored}}{dV_{ac}} = \frac{d(I_{fwd}\tau_p)}{dV_{ac}} = \tau_p \frac{dI_{fwd}}{dV_{ac}} = \tau_p\left(\frac{I_{DC}}{V_t}\right)$$

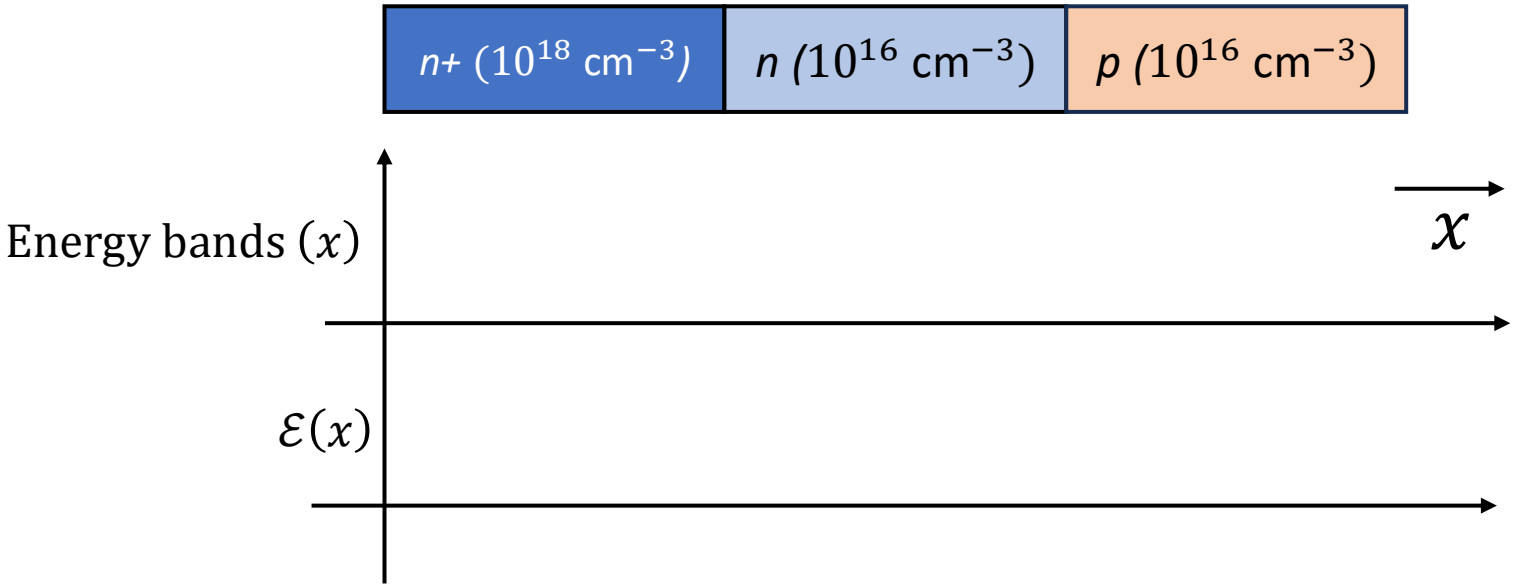
τ_p is minority carrier lifetime in n semiconductor

Since, $C_d \propto I_{DC} \propto e^{\frac{qV_{bias}}{kT}} \gg C_j$ for $V_{bias} > 0.2$ V

Class Test -3 (8 marks) – 20 mins

Consider a silicon $n^+/\mathrm{n}/\mathrm{p}$ diode with structure and doping mentioned in the figure
Given: $n_i = 10^{10}\mathrm{cm}^{-3}$ $\ln(10) = 2.3$, Use Boltzmann carrier statistics, All three regions very long.
(Calculators are not required/allowed)

(a) [3] Sketch qualitatively accurate equilibrium band diagram and electric field across the device ***in the given format***. Label E_c, E_v, E_i, and E_F.



(b) [2] Numerically calculate the built-in potential on the device.

(c) [3] On the same graph, qualitatively sketch and compare the I–V characteristics of a standard $n^+ -$ p diode and given $n^+/\mathrm{n}/\mathrm{p}$ diode. Provide reasoning for differences in (a) I_{rev} and (b) I_{fwd}